Numerical Methods

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Chapter-5 Numerical Methods

5.1 Introduction

- Mathematical models are an integral part in solving engineering problems. Many times, these mathematical models are derived from engineering and science principles, while at other times the models may be obtained from experimental data.
- Mathematical models generally result in need of using mathematical procedures that include but are not limited to

a) Differentiation
b) Nonlinear equations
c) Simultaneous linear equations
d) Curve fitting by interpolation or regression
e) Integration
f) Differential equations

- These mathematical procedures may be suitable to be solved exactly as you must have experienced in the series of calculus courses you have taken, but in most cases, the procedures need to be solved approximately using numerical methods.
- As you know, most differential equations are far too complicated to be solved by an explicit analytic formula. Thus, the development of accurate numerical approximation schemes is essential for both extracting quantitative information as well as achieving a qualitative understanding of the behaviour of their solutions.
- Even in cases, such as the heat and wave equations, where explicit solution formulas (either closed form or infinite series) exist, numerical methods still can be profitably employed. Indeed, one is able to accurately test a proposed numerical algorithm by running it on a known solution. Furthermore, the lessons learned in the design of numerical algorithms for "solved" examples are of inestimable value when confronting more challenging problems.

What you need to know before you start?

1. Calculus
2. Linear Algebra

What the topic covers?

1. Solving Equations by Iterative Methods
   1.1. Bisection Method
   1.2. Newton – Raphson Method
   1.3. Regula Falsi Method
   1.4. Secant Method
2. Numerical solution to Integral equations.
3. Numerical Solution to Differential Equations.

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5.2 Iterative Methods

5.2.1 Bisection Method

After reading this you should be able to:

1. Follow the algorithm of the bisection method of solving a nonlinear equation
2. Use the bisection method to solve examples of finding roots of a nonlinear equation, and
3. Enumerate the advantages and disadvantages of the bisection method.

What is the bisection method and what is it based on?

One of the first numerical methods developed to find the root of a nonlinear equation $f(x)=0$ was the bisection method (also called *binary-search* method). The method is based on the following theorem.

Theorem

An equation $f(x) = 0$, where $f(x)$ is a real continuous function, has at least one root between x_l and x_u if

$f(x_l).\ f(x_u) < 0$ (See Figure 1). Note that if $f(x_l).\ f(x_u) > 0$, there may or may not be any root between x_l and x_u (Figures 2 and 3). If $f(x_l).\ f(x_u) < 0$, then there may be more than one root between x_l and x_u (Figure 4). So the theorem only guarantees one root between x_l and x_u.

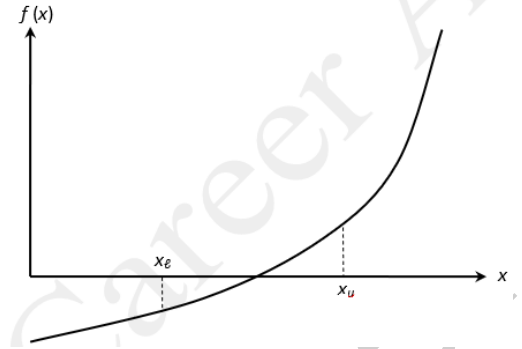
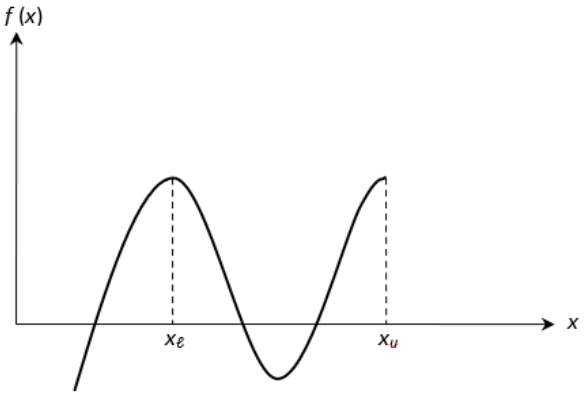


Figure: At least one root exists between the two points if the function is real, continuous, and changessign.

Figure : If the function f(x) does not change sign between the two points, roots of the equation f(x) = 0 may still exist between the two points.



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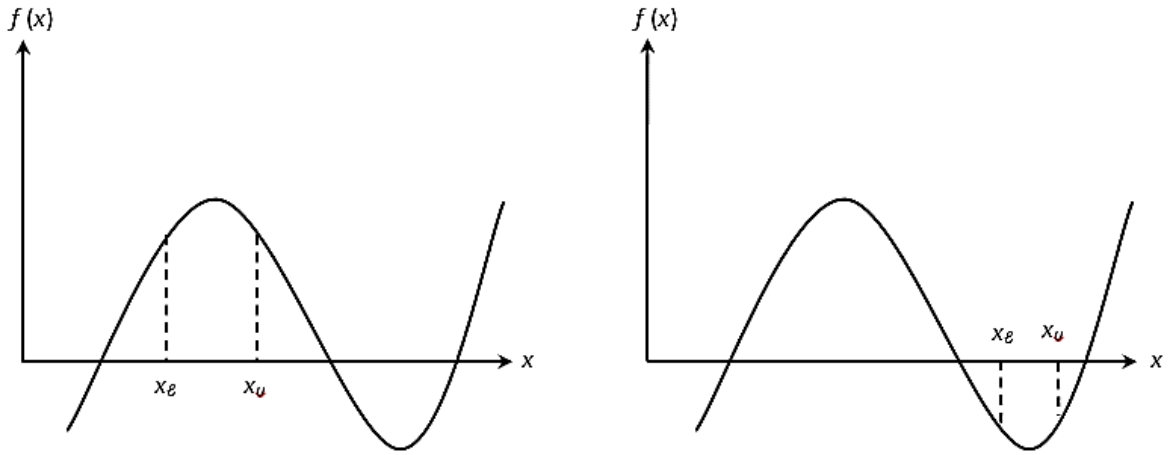


Figure 3: If the function f(x) does not change sign between two points, there may not be any roots for the equation f(x) = 0 between the two points.

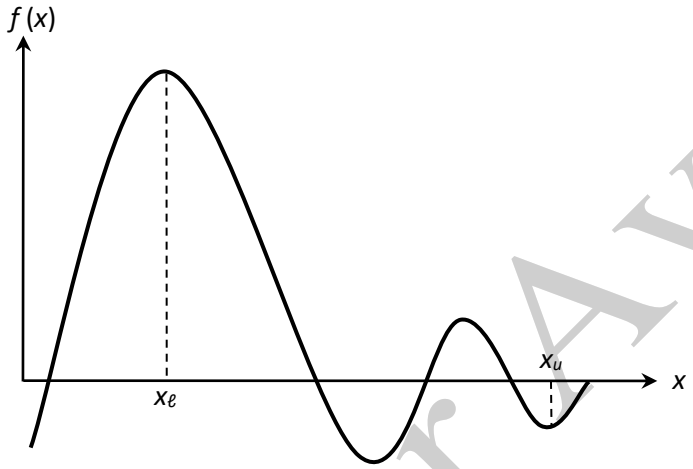


Figure: If the function f(x) changes sign between the two points, more than one root for the equation f(x) = 0 may exist between the two points.

Since the method is based on finding the root between two points, the method falls under the category of bracketing methods.

Since the root is bracketed between two points, x_ℓ and x_u, one can find the mid-point, x_m between x_ℓ and x_u. This gives us two new intervals

1. x_ℓ and x_m, and
2. x_m and x_u.

Is the root now between x_ℓ and x_m or between x_m and x_u? Well, one can find the sign of $f(x_\ell)f(x_m)$, and if $f(x_\ell)f(x_m) < 0$ then the new bracket is between x_ℓ and x_m, otherwise, it is between x_m and x_u. So, you can see that you are literally halving the interval. As one repeats this process, the width of the interval $[x_\ell, x_u]$ becomes smaller and smaller, and you can zero in to the root of the equation $f(x) = 0$. The algorithm for the bisection method is given as follows.

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Algorithm for the bisection method

The steps to apply the bisection method to find the root of the equation f(x) = 0 are

1. Choose x_ℓ and x_u as two guesses for the root such that $f(x_\ell)f(x_u) < 0$, or in other words, $f(x)$ changes sign between x_ℓ and x_u.
2. Estimate the root, x_m, of the equation $f(x) = 0$ as the mid-point between x_ℓ and x_u as
$$x_m = \frac{x_\ell + x_u}{2}$$
3. Now check the following
   a) If $f(x_\ell)f(x_m) < 0$, then the root lies between x_ℓ and x_m; then $x_\ell = x_\ell$ and $x_u = x_m$.
   b) If $f(x_\ell)f(x_m) > 0$, then the root lies between x_m and x_u; then $x_\ell = x_m$ and $x_u = x_u$.
   c) If $f(x_\ell)f(x_m) = 0$; then the root is x_m. Stop the algorithm if this is true.

Advantages of bisection method

a) The bisection method is always convergent. Since the method brackets the root, the method is guaranteed to converge.
b) As iterations are conducted, the interval gets halved. So one can guarantee the error in the solution of the equation.

Drawbacks of bisection method

a) The convergence of the bisection method is slow as it is simply based on halving the interval.
b) If one of the initial guesses is closer to the root, it will take larger number of iterations to reach the root.
c) If a function $f(x)$ is such that it just touches the x-axis such as $f(x) = x^2 = 0$ it will be unable to find the lower guess, x_ℓ, and upper guess, x_u, such that $f(x_\ell)f(x_u) < 0$
d) For functions $f(x)$ where there is a singularity and it reverses sign at the singularity, the bisection method may converge on the singularity. An example includes $f(x) = \frac{1}{x}$

Bisection method converges slowly. It is however, the simplest iterative method and is guaranteed to converge. Convergence is slow and steady. This method cannot be used for finding complex roots.

Question: Use the bisection method of finding roots of equations to find the x of given equation. Conduct three iterations to estimate the root of the below equation which lies between 0 and 0.11.
$x^3 - 0.165x^2 + 3.993\times10^{-4} = 0$

Solution:

Given data

$x_\ell = 0,\ x_u = 0.11$

Check if the function changes sign between x_ℓ and x_u.

$f(x_\ell) = f(0) = (0)^3 - 0.165(0)^2 + 3.993\times10^{-4} = 3.993\times10^{-4}$

$f(x_u) = f(0.11) = (0.11)^3 - 0.165(0.11)^2 + 3.993\times10^{-4} = -2.662\times10^{-4}$

Hence

$f(x_\ell)f(x_u) = f(0)f(0.11) = (3.993\times10^{-4})(-2.662\times10^{-4}) < 0$

So there is at least one root between x_ℓ and x_u, that is between 0 and 0.11.

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Iteration 1

The estimate of the root is

$$x_m = \frac{x_\ell + x_u}{2}$$

$$= \frac{0 + 0.11}{2} = 0.055$$

$f(x_m) = f(0.055) = (0.055)^3 - 0.165(0.055)^2 + 3.993 \times 10^{-4} = 6.655 \times 10^{-5}$

$$f(x_\ell) f(x_u) = f(0) f(0.055) = \left(3.993 \times 10^{-4}\right)\left(6.655 \times 10^{-4}\right) > 0$$

Hence the root is bracketed between x_m and x_u, that is, between 0.055 and 0.11. So, the lower and upper limit of the new bracket is

$x_\ell = 0.055,\ x_u = 0.11$

Iteration 2

The estimate of the root is

$$x_m = \frac{x_\ell + x_u}{2}$$

$$= \frac{0.055 + 0.11}{2} = 0.0825$$

$$f(x_m) = f(0.0825) = (0.0825)^3 - 0.165(0.0825)^2 + 3.993 \times 10^{-4} = -1.622 \times 10^{-4}$$

$$f(x_\ell) f(x_m) = f(0.055) f(0.0825) = \left(6.655 \times 10^{-5}\right) \times \left(-1.622 \times 10^{-4}\right) < 0$$

Hence, the root is bracketed between x_ℓ and x_m, that is, between 0.055 and 0.0825. So the lower and upper limit of the new bracket is $x_\ell = 0.055,\ x_u = 0.0825$

Iteration 3

$$x_m = \frac{x_\ell + x_u}{2}$$

$$= \frac{0.055 + 0.0825}{2} = 0.06875$$

$$f(x_m) = f(0.06875) = (0.06875)^3 - 0.165(0.06875)^2 + 3.993 \times 10^{-4} = -5.563 \times 10^{-5}$$

$$f(x_\ell) f(x_m) = f(0.055) f(0.06875) = (6.655 \times 10^5) \times (-5.563 \times 10^{-5}) < 0$$

Hence, the root is bracketed between x_ℓ and x_m, that is, between 0.055 and 0.06875. So the lower and upper limit of the new bracket is

$x_\ell = 0.055,\ x_u = 0.06875$

As we are supposed to take only three iteration so we will stop here and hence the value of *x* at the end of iteration 3 is 0.06875.

5.2.2 Regula Falsi Method: (Method of False Position)

Like the bisection method, the false position method starts with two points a_0 and b_0 such that $f(a_0)$ and $f(b_0)$ are of opposite signs, which implies by the intermediate value theorem that the function f has a root in the interval $[a_0, b_0]$, assuming continuity of the function f. The method proceeds by producing a sequence of shrinking intervals $[a_k, b_k]$ that all contain a root of f.

Consider $f(x) = 0$,

If $f(a).\ f(b) < 0$ then the iterative formula is

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$$x = \frac{a.f(b) - b.f(a)}{f(b) - f(a)}$$

1. The false position method is guaranteed to converge.
2. It is however, slow as it is first order convergent i.e. order of convergence = 1.
3. It is not necessarily a monotonic convergence to the root but most often it will be superior to Bisection Method

The modified false position will be discussed in later part this chapter.

5.2.3 Newton – Raphson Method

After reading this, you should be able to:

1. Derive the Newton-Raphson method formula,
2. Develop the algorithm of the Newton-Raphson method,
3. Use the Newton-Raphson method to solve a nonlinear equation, and
4. Discuss the drawbacks of the Newton-Raphson method.

Methods such as the bisection method and the false position method of finding roots of a nonlinear equation $f(x) = 0$ require bracketing of the root by two guesses. Such methods are called *bracketing methods*. These methods are always convergent since they are based on reducing the interval between the two guesses so as to zero in on the root of the equation.

In the Newton-Raphson method, the root is not bracketed. In fact, only one initial guess of the root is needed to get the iterative process started to find the root of an equation. The method hence falls in the category of *open methods*. Convergence in open methods is not guaranteed but if the method does converge, it does so much faster than the bracketing methods.

Derivation

The Newton-Raphson method is based on the principle that if the initial guess of the root of $f(x) = 0$ is at x_i, then if one draws the tangent to the curve at $f(x_i)$, the point x_{i+1} where the tangent crosses the x-axis is an improved estimate of the root (Figure 1).

Using the definition of the slope of a function, at $x = x_i$

$$f'(x_i) = \tan\theta$$

$$= \frac{f(x_i) - 0}{x_i - x_{i+1}},$$

which gives

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \qquad (1)$$

Equation (1) is called the Newton-Raphson formula for solving nonlinear equations of the form $f(x) = 0$. So starting with an initial guess, x_i, one can find the next guess, x_{i+1}, by using Equation (1). One can repeat this process until one finds the root within a desirable tolerance.

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Algorithm:

The steps of the Newton-Raphson method to find the root of an equation $f(x)=0$ are

1. Evaluate $f'(x)$ symbolically
2. Use an initial guess of the root, x_i, to estimate the new value of the root, x_{i+1}, as

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

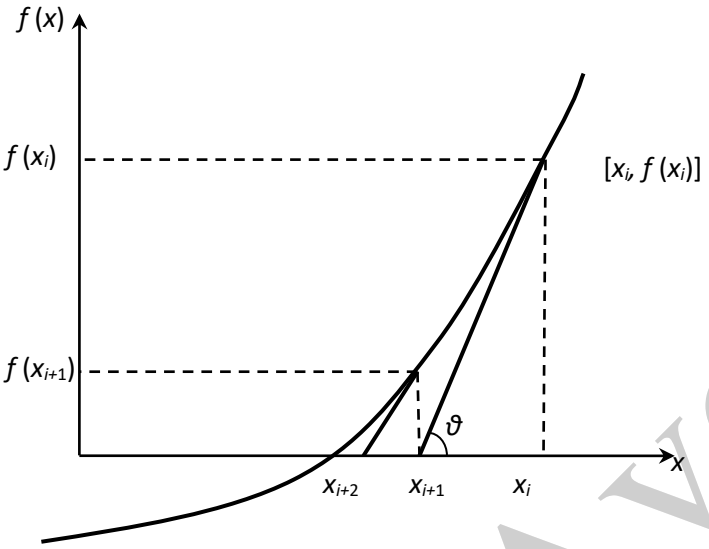


Figure: Geometrical illustration of the Newton-Raphson method.

Drawbacks of Newton Raphson Method

1. Divergence at inflection points
2. Division by zero condition is not allowed
3. Oscillations occur near local minimum and maximum.

Points to Remember:

1. Newton Raphson method is useful in cases of large value of *f* '(x) i.e. when the graph of *f* (x) while crossing the x – axis vertical.
2. Sensitive to starting value. Convergence is fast if starting point near the root i.e. the newton's formula converges provided the initial approximation x_0 is chosen sufficiently close to the root.
3. Newton's method is generally used to improve result obtained by other methods
4. Newton's method has quadratic convergence. i.e. order of convergence is = 2 i.e. the subsequent error at each step is proportional to the square of the error at previous step.
5. The number of functions to be evaluated per iteration is 2.

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Question: Use the Newton - Raphson method of finding roots of equations to find the x of given equation. Conduct three iterations to estimate the root of the below equation which lies nearer to 0.05.

$x^3 - 0.165x^2 + 3.993\times10^{-4} = 0$

Solution:

$f(x) = x^3 - 0.165x^2 + 3.993\times10^{-4}$

$f'(x) = 3x^2 - 0.33x$

The initial root is $x_0 = 0.05$

Iteration 1

The estimate of the root is

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 0.05 - \frac{(0.05)^3 - 0.165(0.05)^2 + 3.993\times10^{-4}}{3(0.05)^2 - 0.33(0.05)}$$

$$= 0.05 - \frac{1.118\times10^{-4}}{-9\times10^{-3}}$$

$$= 0.05 - (-0.01242) \quad = 0.06242$$

Iteration 2

The estimate of the root is

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 0.06242 - \frac{(0.06242)^3 - 0.165(0.06242)^2 + 3.993\times10^{-4}}{3(0.06242)^2 - 0.33(0.06242)}$$

$$= 0.06242 - \frac{-3.97781\times10^{-7}}{-8.90973\times10^{-3}}$$

$$= 0.06242 - \left(4.4646\times10^{-5}\right)$$

$$= 0.06238$$

Iteration 3

The estimate of the root is

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 0.06238 - \frac{(0.06238)^3 - 0.165(0.06238)^2 + 3.993\times10^{-4}}{3(0.06238)^2 - 0.33(0.06238)}$$

$$= 0.06238 - \frac{4.44\times10^{-11}}{-8.91171\times10^{-3}}$$

$$= 0.06238 - \left(-4.9822\times10^{-9}\right)$$

$$= 0.06238$$

So the value of x at the end of third iteration is 0.06238

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5.2.4 Secant Method (Modified false position method)

After reading this, you should be able to:

1. Derive the secant method to solve for the roots of a nonlinear equation,
2. Use the secant method to numerically solve a nonlinear equation.

What is the secant method and why would I want to use it instead of the Newton-Raphson method?

The Newton - Raphson method of solving a nonlinear equation $f(x)=0$ is given by the iterative formula

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \qquad (1)$$

One of the drawbacks of the Newton-Raphson method is that you have to evaluate the derivative of the function. To overcome these drawbacks, the derivative of the function, $f(x)$ is approximated as

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \qquad (2)$$

Substituting Equation (2) in Equation (1) gives

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})} \qquad (3)$$

The above equation is called the secant method. This method now requires two initial guesses, but unlike the bisection method, the two initial guesses do not need to bracket the root of the equation. The secant method is an open method and may or may not converge. However, when secant method converges, it will typically converge faster than the bisection method. However, since the derivative is approximated as given by Equation (2), it typically converges slower than the Newton-Raphson method.

The secant method can also be derived from geometry, as shown in Figure 1. Taking two initial guesses, x_{i-1} and x_i, one draws a straight line between $f(x_i)$ and $f(x_{i-1})$ passing through the x-axis at x_{i+1}. *ABE* and *DCE* are similar triangles.

Hence

$$\frac{AB}{AE} = \frac{DC}{DE}$$

$$\frac{f(x_i)}{x_i - x_{i+1}} = \frac{f(x_{i-1})}{x_{i-1} - x_{i+1}}$$

On rearranging, the secant method is given as

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

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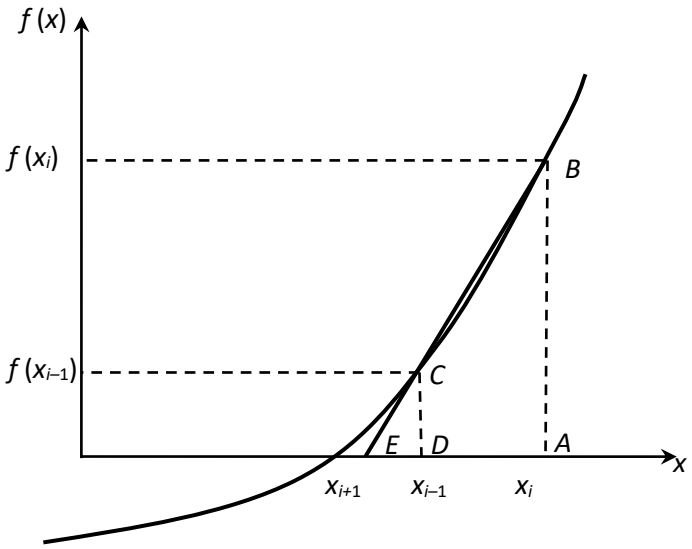


Figure: Geometrical representation of the secant method

Points to remember:

1. In this method, instead of choosing two values of x such that the function has opposite signs at these values we choose two values nearest to the root.
2. This method is applicable only if one is sure that there is root in the vicinity of x_1 the starting value.
3. On the average, Secant method is more efficient compared to Newton Raphson Method.
4. In this method, we approximate the graph of the function y = f(x) in the neighborhood of the root by a straight line (secant passing through the points (x_{i-1} , f_{i-1}) and (x_i , f_i).
5. The order of convergence is 1.62, i.e. converges faster than false position method.
6. No guarantee of convergence if not near root. The method fails if f (x_i) = f (x_{i-1}).
7. The amount of computational effort is one function evaluation.

Question:

Use the Secant method of finding roots of equations to find the x of given equation. Conduct three iterations to estimate the root of the below equation which lies nearer to 0.05.

$$x^3 - 0.165x^2 + 3.993\times10^{-4} = 0$$

Solution:

Iteration 1

The estimate of the root is

$$x_1 = x_0 - \frac{f(x_0)(x_0 - x_{-1})}{f(x_0) - f(x_{-1})}$$

$$= x_0 - \frac{\left(x_0^3 - 0.165x_0^2 + 3.993\times10^{-4}\right)\times\left(x_0 - x_{-1}\right)}{\left(x_0^3 - 0.165x_0^2 + 3.993\times10^{-4}\right) - \left(x_{-1}^3 - 0.165x_{-1}^2 + 3.993\times10^{-4}\right)}$$

$$= 0.05 - \frac{\left[0.05^3 - 0.165(0.05)^2 + 3.993\times10^{-4}\right]\times[0.05 - 0.02]}{\left[0.05^3 - 0.165(0.05)^2 + 3.993\times10^{-4}\right] - \left[0.02^3 - 0.165(0.02)^2 + 3.993\times10^{-4}\right]} = 0.06461$$

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Iteration 2

$$x_2 = x_1 - \frac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$= x_1 - \frac{\left(x_1^3 - 0.165x_1^2 + 3.993\times10^{-4}\right)\times(x_1 - x_0)}{\left(x_1^3 - 0.165x_1^2 + 3.993\times10^{-4}\right) - \left(x_0^3 - 0.165x_0^2 + 3.993\times10^{-4}\right)}$$

$$= 0.06461 - \frac{\left[0.06461^3 - 0.165(0.06461)^2 + 3.993\times10^{-4}\right]\times(0.06461 - 0.05)}{\left[0.06461^3 - 0.165(0.06461)^2 + 3.993\times10^{-4}\right] - \left[0.05^3 - 0.165(0.05)^2 + 3.993\times10^{-4}\right]}$$

$$= 0.06241$$

Iteration 3

$$x_3 = x_2 - \frac{f(x_2)(x_2 - x_1)}{f(x_2) - f(x_1)}$$

$$= x_2 - \frac{\left(x_2^3 - 0.165x_2^2 + 3.993\times10^{-4}\right)\times(x_2 - x_1)}{\left(x_2^3 - 0.165x_2^2 + 3.993\times10^{-4}\right) - \left(x_1^3 - 0.165x_1^2 + 3.993\times10^{-4}\right)}$$

$$= 0.06241 - \frac{\left[0.06241^3 - 0.165(0.06241)^2 + 3.993\times10^{-4}\right]\times(0.06241 - 0.06461)}{\left[0.06241^3 - 0.165(0.06241)^2 + 3.993\times10^{-4}\right] - \left[0.06461^3 - 0.165(0.06461)^2 + 3.993\times10^{-4}\right]}$$

$$= 0.06238$$

Therefore, the value of root after three iteration will be 0.06238

Do-it-yourself (DIY exercise)
1.Use your calculator and find out the correct root of the equation used in above solved examples. **2.Which method from all methods discussed above is more correct?**

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5.3 Numerical Integration

5.3.1 Trapezoidal Rule

After reading this, you should be able to:

1. derive the trapezoidal rule of integration,
2. use the trapezoidal rule of integration to solve problems,
3. derive the multiple-segment trapezoidal rule of integration,
4. use the multiple-segment trapezoidal rule of integration to solve problems.

The dictionary definition of *integration* is combining parts so that they work together or form a whole. Mathematically, integration stands for finding the area under a curve from one point to another. It is represented by

$$\int_a^b f(x)dx$$

where the symbol $\int$ is an integral sign, and a and b are the lower and upper limits of integration, respectively, the function f is the integrand of the integral, and x is the variable of integration.

Here, we will discuss the trapezoidal rule of approximating integrals of the form

$$I = \int_a^b f(x)dx$$

where

$f(x)$ is called the integrand,
$a =$ lower limit of integration
$b =$ upper limit of integration

What is the trapezoidal rule?

The trapezoidal rule is based on the Newton-Cotes formula that if one approximates the integrand by an nth order polynomial, then the integral of the function is approximated by the integral of that n^{th} order polynomial.

Integrating polynomials is simple and is based on the calculus formula.

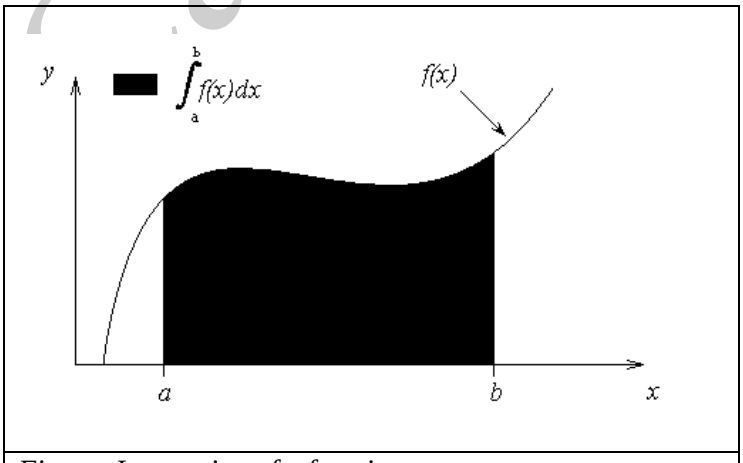


Figure: Integration of a function

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$$\int_a^b x^n dx = \left(\frac{b^{n+1} - a^{n+1}}{n+1} \right), \; n \neq -1 \tag{1}$$

So if we want to approximate the integral

$$I = \int_a^b f(x)dx \tag{2}$$

to find the value of the above integral, one assumes

$$f(x) \approx f_n(x) \tag{3}$$

where

$$f_n(x) = a_0 + a_1 x + ... + a_{n-1} x^{n-1} + a_n x^n \, . \tag{4}$$

where $f_n(x)$ is a nth order polynomial. The trapezoidal rule assumes $n = 1$, that is, approximating the integral by a linear polynomial (straight line),

$$\int_a^b f(x)dx \approx \int_a^b f_1(x)dx$$

Multiple-Segment Trapezoidal Rule

Divide $(b-a)$ into n equal segments as shown in Figure 4. Then the width of each segment is

$$h = \frac{b-a}{n} \tag{13}$$

The integral I can be broken into h integrals as

$$I = \int_a^b f(x)dx$$

$$= \int_a^{a+h} f(x)dx + \int_{a+h}^{a+2h} f(x)dx + ... + \int_{a+(n-2)h}^{a+(n-1)h} f(x)dx + \int_{a+(n-1)h}^{b} f(x)dx \tag{14}$$

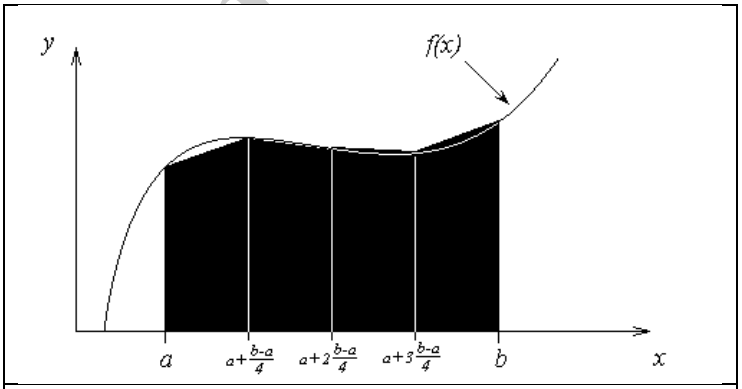


Figure: Multiple ($n = 4$) segment trapezoidal rule

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Applying trapezoidal rule on each segment gives

$$\int_a^b f(x)dx = [(a+h)-a]\left[\frac{f(a)+f(a+h)}{2}\right]$$

$$+[(a+2h)-(a+h)]\left[\frac{f(a+h)+f(a+2h)}{2}\right]$$

$$+ \ldots\ldots\ldots\ldots + [(a+(n-1)h)-(a+(n-2)h)]\left[\frac{f(a+(n-2)h)+f(a+(n-1)h)}{2}\right]$$

$$+[b-(a+(n-1)h)]\left[\frac{f(a+(n-1)h)+f(b)}{2}\right]$$

$$= h\left[\frac{f(a)+f(a+h)}{2}\right] + h\left[\frac{f(a+h)+f(a+2h)}{2}\right] + \ldots\ldots\ldots\ldots$$

$$+h\left[\frac{f(a+(n-2)h)+f(a+(n-1)h)}{2}\right] + h\left[\frac{f(a+(n-1)h)+f(b)}{2}\right]$$

$$= h\left[\frac{f(a)+2f(a+h)+2f(a+2h)+\ldots+2f(a+(n-1)h)+f(b)}{2}\right]$$

$$= \frac{h}{2}\left[f(a)+2\left\{\sum_{i=1}^{n-1} f(a+ih)\right\}+f(b)\right]$$

$$= \frac{b-a}{2n}\left[f(a)+2\left\{\sum_{i=1}^{n-1} f(a+ih)\right\}+f(b)\right] \qquad (15)$$

Trapezoidal Rule:

Let $y = f(x)$ be a function defined on $[a,b]$ which is subdivided into n equal sub-intervals each of width h, so that $b-a=nh$, then

$$\int_{x_o}^{x_o+nh} f(x)dx = \frac{h}{2}\left[(y_0+y_n)+2(y_1+y_2+y_3+\ldots.+y_{n-1})\right]$$

where n is any positive integer and y_n is the value of $f(x)$ for $x = a+nh$.

i.e $y_0 = f(a), y_1 = f(a+h), y_2 = f(a+2h), \ldots.. y_n = f(a+nh) = f(b)$

Question:

The vertical distance covered by a rocket from $t=8$ to $t=30$ seconds is given by

$$x = \int_8^{30}\left(2000\ln\left[\frac{140000}{140000-2100t}\right]-9.8t\right)dt$$

Use the two-segment trapezoidal rule to find the distance covered from $t=8$ to $t=30$ seconds.

Solution:

The solution using 2-segment Trapezoidal rule is

$$I \approx \frac{b-a}{2n}\left[f(a)+2\left\{\sum_{i=1}^{n-1} f(a+ih)\right\}+f(b)\right]$$

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$n = 2$

$a = 8$

$b = 30$

$$h = \frac{b-a}{n}$$

$$= \frac{30-8}{2} = 11$$

$$I \approx \frac{30-8}{2(2)}\left[f(8) + 2\left\{ \sum_{i=1}^{2-1} f(8+11i) \right\} + f(30) \right]$$

$$= \frac{22}{4}\left[f(8) + 2f(19) + f(30) \right] = \frac{22}{4}\left[177.27 + 2(484.75) + 901.67 \right]$$

$$= 11266 \text{ m}$$

5.3.2 Simpson's 1/3rd Rule

After reading this, you should be able to

1. Derive the formula for Simpson's 1/3 rule of integration,
2. Use Simpson's 1/3 rule it to solve integrals,
3. Develop the formula for multiple-segment Simpson's 1/3 rule of integration,
4. Use multiple-segment Simpson's 1/3 rule of integration to solve integrals, and
5. Derive the true error formula for multiple-segment Simpson's 1/3 rule.

Sometimes, the evaluation of expressions involving these integrals can become daunting, if not indeterminate. For this reason, a wide variety of numerical methods has been developed to simplify the integral. Here, we will discuss Simpson's 1/3 rule of integral approximation, which improves upon the accuracy of the trapezoidal rule.

Here, we will discuss the Simpson's 1/3 rule of approximating integrals of the form

$$I = \int_a^b f(x)\,dx$$

where

$f(x)$ is called the integrand,

$a =$ Lower limit of integration

$b =$ Upper limit of integration

The trapezoidal rule was based on approximating the integrand by a first order polynomial, and then integrating the polynomial over interval of integration. Simpson's 1/3 rule is an extension of Trapezoidal rule where the integrand is approximated by a second order polynomial.

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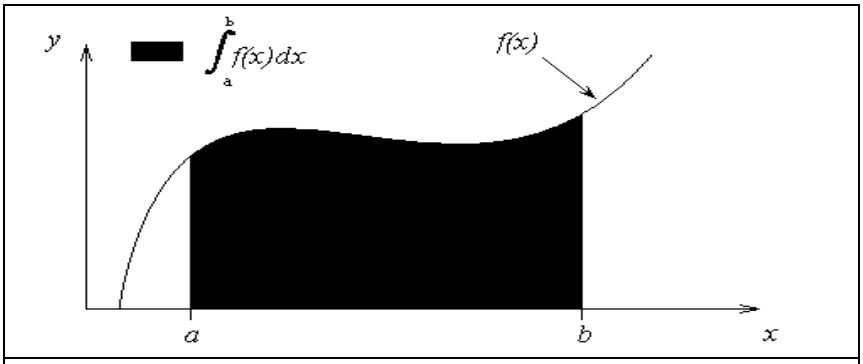


Figure - Integration of a function

Multiple-segment Simpson's 1/3 Rule

Just like in multiple-segment trapezoidal rule, one can subdivide the interval $[a,b]$ into n segments and apply Simpson's 1/3 rule repeatedly over every two segments. **Note that n needs to be even.** Divide interval $[a,b]$ into n equal segments, so that the segment width is given by

$$h = \frac{b-a}{n}.$$

Now

$$\int_a^b f(x)dx = \int_{x_0}^{x_n} f(x)dx$$

where

$$x_0 = a$$

$$x_n = b$$

$$\int_a^b f(x)dx = \int_{x_0}^{x_2} f(x)dx + \int_{x_2}^{x_4} f(x)dx + ... + \int_{x_{n-4}}^{x_{n-2}} f(x)dx + \int_{x_{n-2}}^{x_n} f(x)dx$$

Apply Simpson's 1/3rd Rule over each interval,

$$\int_a^b f(x)dx \cong (x_2 - x_0)\left[\frac{f(x_0) + 4f(x_1) + f(x_2)}{6}\right] + (x_4 - x_2)\left[\frac{f(x_2) + 4f(x_3) + f(x_4)}{6}\right] + ...$$

$$+ (x_{n-2} - x_{n-4})\left[\frac{f(x_{n-4}) + 4f(x_{n-3}) + f(x_{n-2})}{6}\right] + (x_n - x_{n-2})\left[\frac{f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)}{6}\right]$$

Since

$$x_i - x_{i-2} = 2h$$

$$i = 2,\ 4,\ ...,\ n$$

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then

$$\int_a^b f(x)dx \cong 2h\left[\frac{f(x_0)+4f(x_1)+f(x_2)}{6}\right]+2h\left[\frac{f(x_2)+4f(x_3)+f(x_4)}{6}\right]+\ldots$$

$$+2h\left[\frac{f(x_{n-4})+4f(x_{n-3})+f(x_{n-2})}{6}\right]+2h\left[\frac{f(x_{n-2})+4f(x_{n-1})+f(x_n)}{6}\right]$$

$$=\frac{h}{3}\left[f(x_0)+4\left\{f(x_1)+f(x_3)+\ldots+f(x_{n-1})\right\}+2\left\{f(x_2)+f(x_4)+\ldots+f(x_{n-2})\right\}+f(x_n)\right]$$

$$=\frac{h}{3}\left[f(x_0)+4\sum_{\substack{i=1\\ i=odd}}^{n-1}f(x_i)+2\sum_{\substack{i=2\\ i=even}}^{n-2}f(x_i)+f(x_n)\right]$$

$$\boxed{\int_a^b f(x)dx \cong \frac{b-a}{3n}\left[f(x_0)+4\sum_{\substack{i=1\\ i=odd}}^{n-1}f(x_i)+2\sum_{\substack{i=2\\ i=even}}^{n-2}f(x_i)+f(x_n)\right]}$$

Simpson's One Third Rule:-

Let $y=f(x)$ be a function defined on $[a,b]$ which is subdivided into n equal sub-intervals ($n=even\,number$) each of width h, so that $b-a=nh$, then

$$\int_{x_o}^{x_o+nh} f(x)dx=\frac{h}{3}\left[(y_0+y_n)+4(y_1+y_3+\ldots.+y_{n-1})+2(y_2+y_4+\ldots.+y_{n-2})\right]$$

where n is any positive integer and y_n is the value of $f(x)$ for $x=a+nh$.

i.e $y_0=f(a), y_1=f(a+h), y_2=f(a+2h),\ldots..y_n=f(a+nh)=f(b)$

Question:

Use 4-segment Simpson's 1/3 rule to approximate the distance covered by a rocket in meters from $t=8$ s to $t=30$ s as given by

$$x=\int_8^{30}\left(2000\ln\left[\frac{140000}{140000-2100t}\right]-9.8t\right)dt$$

a) Use four segment Simpson's 1/3rd Rule to find the probability.

Solution:

a) Using n segment Simpson's 1/3 rule,

$$x\approx\frac{b-a}{3n}\left[f(t_0)+4\sum_{\substack{i=1\\ i=odd}}^{n-1}f(t_i)+2\sum_{\substack{i=2\\ i=even}}^{n-2}f(t_i)+f(t_n)\right]$$

$n=4$

$a=8$

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$b = 30$

$h = \dfrac{b-a}{n}$

$= \dfrac{30-8}{4}$

$= 5.5$

$f(t) = 2000\ln\left[\dfrac{140000}{140000-2100t}\right] - 9.8t$

So

$f(t_0) = f(8)$

$f(8) = 2000\ln\left[\dfrac{140000}{140000-2100(8)}\right] - 9.8(8) = 177.27 m/s$

$f(t_1) = f(8+5.5) = f(13.5)$

$f(13.5) = 2000\ln\left[\dfrac{140000}{140000-2100(13.5)}\right] - 9.8(13.5) = 320.25 m/s$

$f(t_2) = f(13.5+5.5) = f(19)$

$f(19) = 2000\ln\left(\dfrac{140000}{140000-2100(19)}\right) - 9.8(19) = 484.75 m/s$

$f(t_3) = f(19+5.5) = f(24.5)$

$f(24.5) = 2000\ln\left[\dfrac{140000}{140000-2100(24.5)}\right] - 9.8(24.5) = 676.05 m/s$

$f(t_4) = f(t_n) = f(30)$

$f(30) = 2000\ln\left[\dfrac{140000}{140000-2100(30)}\right] - 9.8(30) = 901.67 m/s$

$$x = \frac{b-a}{3n}\left[f(t_0) + 4\sum_{\substack{i=1 \\ i=odd}}^{n-1} f(t_i) + 2\sum_{\substack{i=2 \\ i=even}}^{n-2} f(t_i) + f(t_n)\right]$$

$$= \frac{30-8}{3(4)}\left[f(8) + 4\sum_{\substack{i=1 \\ i=odd}}^{3} f(t_i) + 2\sum_{\substack{i=2 \\ i=even}}^{2} f(t_i) + f(30)\right]$$

$$= \frac{22}{12}\left[f(8) + 4f(t_1) + 4f(t_3) + 2f(t_2) + f(30)\right]$$

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$$= \frac{11}{6}\left[f(8) + 4f(13.5) + 4f(24.5) + 2f(19) + f(30)\right]$$
$$= \frac{11}{6}\left[177.27 + 4(320.25) + 4(676.05) + 2(484.75) + 901.67\right]$$
$$= 11061.64\ m$$

5.4 Numerical solution to ordinary differential equations:

An equation that consists of derivatives is called a differential equation. Differential equations have applications in all areas of science and engineering. Mathematical formulation of most of the physical and engineering problems leads to differential equations. So, it is important for engineers and scientists to know how to set up differential equations and solve them.

5.4.1 Euler's Method:

After reading this topic, you should be able to:

1. Develop Euler's Method for solving ordinary differential equations,
2. Determine how the step size affects the accuracy of a solution,
3. Derive Euler's formula from Taylor series, and
4. Use Euler's method to find approximate values of integrals.

Euler's method is a numerical technique to solve ordinary differential equations of the form

$$\frac{dy}{dx} = f(x, y), y(0) = y_0 \qquad (1)$$

So only first order ordinary differential equations can be solved by using Euler's method. In another chapter we will discuss how Euler's method is used to solve higher order ordinary differential equations or coupled (simultaneous) differential equations. How does one write a first order differential equation in the above form?

Derivation of Euler's method:

At $x = 0$, we are given the value of $y = y_0$. Let us call $x = 0$ as x_0. Now since we know the slope of y with respect to x, that is, $f(x, y)$, then at $x = x_0$, the slope is $f(x_0, y_0)$. Both x_0 and y_0 are known from the initial condition $y(x_0) = y_0$.

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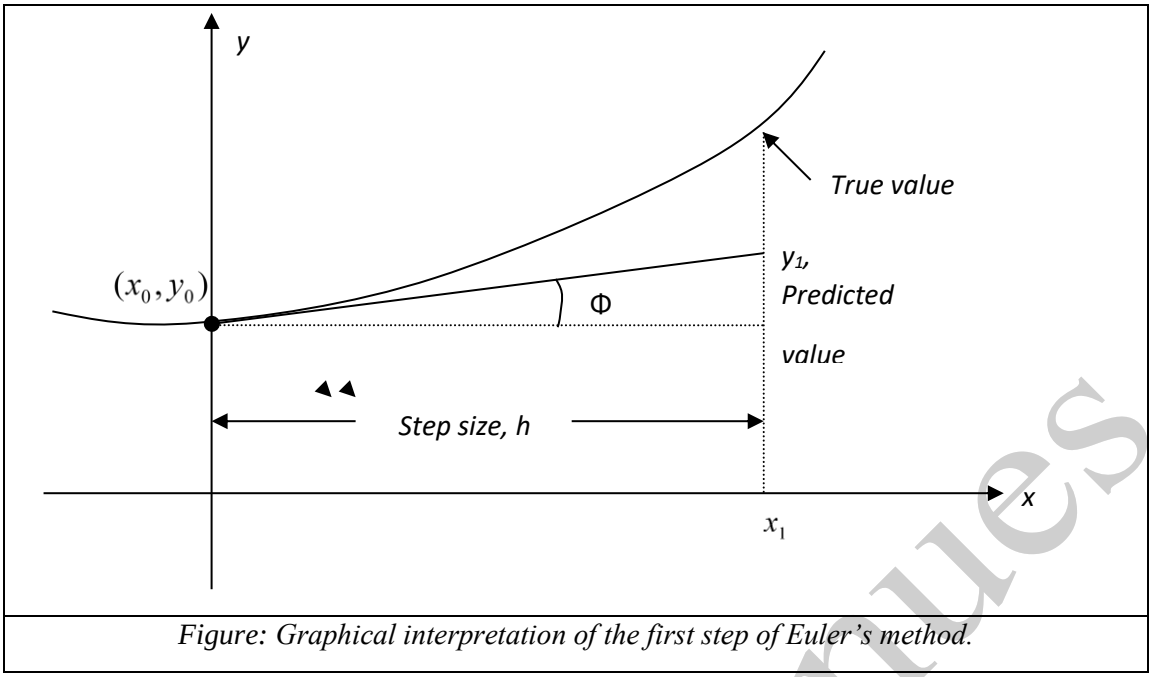


Figure: Graphical interpretation of the first step of Euler's method.

So, the slope at $x = x_0$ as shown in Figure 1 is

$$\text{Slope} = \frac{Rise}{Run}$$

$$= \frac{y_1 - y_0}{x_1 - x_0}$$

$$= f(x_0, y_0)$$

From here

$$y_1 = y_0 + f(x_0, y_0)(x_1 - x_0)$$

Calling $x_1 - x_0$ the step size h, we get

$$y_1 = y_0 + f(x_0, y_0)h \qquad (2)$$

One can now use the value of y_1 (an approximate value of y at $x = x_1$) to calculate y_2, and that would be the predicted value at x_2, given by

$$y_2 = y_1 + f(x_1, y_1)h$$

$$x_2 = x_1 + h$$

Based on the above equations, if we now know the value of $y = y_i$ at x_i, then

$$y_{i+1} = y_i + f(x_i, y_i)h \qquad (3)$$

This formula is known as Euler's method and is illustrated graphically in Figure 2. In some books, it is also called the Euler-Cauchy method.

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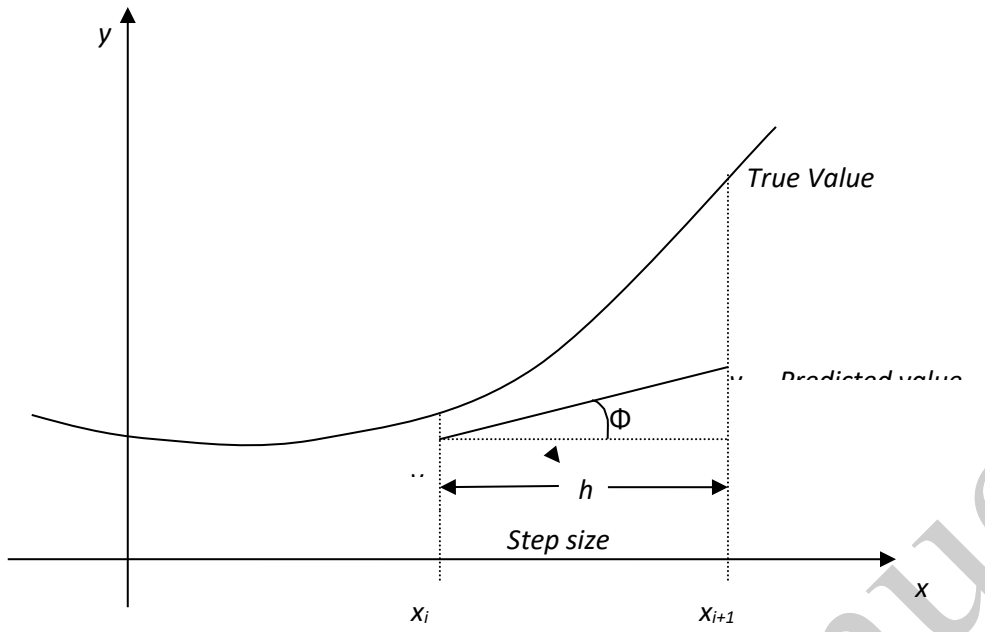


Figure: General graphical interpretation of Euler's method.

Euler's Method:

Let $\frac{dy}{dx} = f(x,y)$ and $y = y_0$ *at* $x = x_0$

The formula for the solution of $\frac{dy}{dx} = f(x,y)$ by Euler's Iterative Method for n[th] iteration is

$$y_{n+1} = y_n + h\, f(x_n, y_n)$$

$$y_1 = y_0 + h\, f(x_0, y_0)$$

$$y_2 = y_1 + h\, f(x_1, y_1)$$

$$y_3 = y_2 + h\, f(x_2, y_2)$$

.....

.....

and so on

$$y_0 = f(a) = f(x_0),$$

Where $y_1 = f(a+h) = f(x_1)$, $y_2 = f(a+2h) = f(x_2), \ldots..$

and

$$y_n = f(a+nh) = f(b)$$

Taylor's Series Method:

Let $\frac{dy}{dx} = f(x,y)$ and $y = y_0$ *at* $x = x_0 = a$ or $y(x_0) = value\, of\ y\ is\ y_0\ at\ x = x_0$

The formula for the solution of $\frac{dy}{dx} = f(x,y)$ by Taylor's series at $x = a + h$ is

$$y(a+h) = y_0 + h\,\dot{f}(a) + \frac{h^2}{2!}\,\ddot{f}(a) + \ldots..$$

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$$where\ \dot{f}(a) = \frac{dy}{dx} \text{ at } x = x_0, and\ y = y_0\ ,$$

$$\ddot{f}(a) = \frac{d^2y}{dx^2} \text{ at } x = x_0, and\ y = y_0$$

$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

OR

$$y(x_0 + h) = y(x_0) + \frac{h}{1!}\dot{y}(x_0) + \frac{h}{2!}\ddot{y}(x_0) + \ldots\ldots$$

$$where\ \dot{y}(x_0) = \frac{dy}{dx} \text{ at } x = x_0, and\ y = y_0\ ,$$

$$\ddot{y}(x_0) = \frac{d^2y}{dx^2} \text{ at } x = x_0, and\ y = y_0$$

$$\ldots\ldots\ldots\ldots\ldots$$

Points to Remember:

1. It is based on the linear term in Taylor series expansion of $f(x, y)$
2. The error build using this method is substantial hence it is rarely used.
3. This method is also called Runge Kutta first order method.

Question:

Find an approximate value of

$$\int_5^8 6x^3 dx$$

using Euler's method of solving an ordinary differential equation. Use a step size of $h = 1.5$.

Solution:

Given $\int_5^8 6x^3 dx$, we can rewrite the integral as the solution of an ordinary differential equation

$$\frac{dy}{dx} = 6x^3,\ y(5) = 0$$

where $y(8)$ will give the value of the integral $\int_5^8 6x^3 dx$.

$$\frac{dy}{dx} = 6x^3 = f(x, y),\ y(5) = 0$$

The Euler's method equation is

$$y_{i+1} = y_i + f(x_i, y_i)h$$

Step 1

$$i = 0,\ x_0 = 5,\ y_0 = 0$$

$h = 1.5$

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$$x_1 = x_0 + h$$
$$= 5 + 1.5$$
$$= 6.5$$
$$y_1 = y_0 + f(x_0, y_0)h$$
$$= 0 + f(5,0) \times 1.5$$
$$= 0 + (6 \times 5^3) \times 1.5$$
$$= 1125$$
$$\approx y(6.5)$$

Step 2

$$i = 1,\ x_1 = 6.5,\ y_1 = 1125$$
$$x_2 = x_1 + h$$
$$= 6.5 + 1.5$$
$$= 8$$
$$y_2 = y_1 + f(x_1, y_1)h$$
$$= 1125 + f(6.5,1125) \times 1.5$$
$$= 1125 + (6 \times 6.5^3) \times 1.5$$
$$= 3596.625$$
$$\approx y(8)$$

Hence

$$\int_5^8 6x^3 dx = y(8) - y(5)$$
$$\approx 3596.625 - 0 \quad = 3596.625$$

5.4.2 Runge Kutta 2nd order Method:

After reading this , you should be able to:

1. Understand the Runge-Kutta 2nd order method for ordinary differential equations and how to use it to solve problems.

What is the Runge-Kutta 2nd order method?

The Runge-Kutta 2nd order method is a numerical technique used to solve an ordinary differential equation of the form

$$\frac{dy}{dx} = f(x, y), y(0) = y_0$$

Euler's method is given by

$$y_{i+1} = y_i + f(x_i, y_i)h \qquad (1)$$

where

$$x_0 = 0$$

$$y_0 = y(x_0)$$

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$$h = x_{i+1} - x_i$$

To understand the Runge-Kutta 2nd order method, we need to derive Euler's method from the Taylor series.

$$y_{i+1} = y_i + \left.\frac{dy}{dx}\right|_{x_i, y_i} (x_{i+1} - x_i) + \frac{1}{2!}\left.\frac{d^2y}{dx^2}\right|_{x_i, y_i} (x_{i+1} - x_i)^2 + \frac{1}{3!}\left.\frac{d^3y}{dx^3}\right|_{x_i, y_i} (x_{i+1} - x_i)^3 + \ldots$$

$$= y_i + f(x_i, y_i)(x_{i+1} - x_i) + \frac{1}{2!} f'(x_i, y_i)(x_{i+1} - x_i)^2 + \frac{1}{3!} f''(x_i, y_i)(x_{i+1} - x_i)^3 + \ldots \quad (2)$$

As you can see the first two terms of the Taylor series

$$y_{i+1} = y_i + f(x_i, y_i) h$$

are Euler's method and hence can be considered to be the Runge - Kutta 1st order method.

The true error in the approximation is given by

$$E_t = \frac{f'(x_i, y_i)}{2!} h^2 + \frac{f''(x_i, y_i)}{3!} h^3 + \ldots \quad (3)$$

So what would a 2nd order method formula look like. It would include one more term of the Taylor series as follows.

$$y_{i+1} = y_i + f(x_i, y_i) h + \frac{1}{2!} f'(x_i, y_i) h^2 \quad (4)$$

Let us take a generic example of a first order ordinary differential equation

$$\frac{dy}{dx} = e^{-2x} - 3y, y(0) = 5$$

$$f(x, y) = e^{-2x} - 3y$$

Now since y is a function of x,

$$f'(x, y) = \frac{\partial f(x, y)}{\partial x} + \frac{\partial f(x, y)}{\partial y}\frac{dy}{dx} \quad (5)$$

$$= \frac{\partial}{\partial x}\left(e^{-2x} - 3y\right) + \frac{\partial}{\partial y}\left[\left(e^{-2x} - 3y\right)\right]\left(e^{-2x} - 3y\right)$$

$$= -2e^{-2x} + (-3)\left(e^{-2x} - 3y\right)$$

$$= -5e^{-2x} + 9y$$

The 2nd order formula for the above example would be

$$y_{i+1} = y_i + f(x_i, y_i) h + \frac{1}{2!} f'(x_i, y_i) h^2$$

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$$= y_i + \left(e^{-2x_i} - 3y_i\right)h + \frac{1}{2!}\left(-5e^{-2x_i} + 9y_i\right)h^2$$

However, we already see the difficulty of having to find $f'(x, y)$ in the above method. What Runge and Kutta did was write the 2nd order method as

$$y_{i+1} = y_i + \left(a_1 k_1 + a_2 k_2\right)h \qquad (6)$$

where

$$k_1 = f\left(x_i, y_i\right)$$

$$k_2 = f\left(x_i + p_1 h, y_i + q_{11} k_1 h\right) \qquad (7)$$

This form allows one to take advantage of the 2nd order method without having to calculate $f'(x, y)$.

So how do we find the unknowns a_1, a_2, p_1 and q_{11}. Without proof (see Appendix for proof), equating Equation (4) and (6) , gives three equations.

$$a_1 + a_2 = 1$$

$$a_2 p_1 = \frac{1}{2}$$

$$a_2 q_{11} = \frac{1}{2}$$

Since we have 3 equations and 4 unknowns, we can assume the value of one of the unknowns. The other three will then be determined from the three equations. Generally, the value of a_2 is chosen to evaluate the other three constants. The three values generally used for a_2 are $\frac{1}{2}$, 1 and $\frac{2}{3}$, and are known as Heun's Method, the midpoint method and Ralston's method, respectively.

Heun's Method

Here $a_2 = \frac{1}{2}$ is chosen, giving

$$a_1 = \frac{1}{2}$$

$$p_1 = 1$$

$$q_{11} = 1$$

resulting in

$$y_{i+1} = y_i + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right)h \quad (8)$$

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where

$$k_1 = f\left(x_i, y_i\right) \quad (9a)$$

$$k_2 = f\left(x_i + h, y_i + k_1 h\right) \quad (9b)$$

This method is graphically explained in Figure 1.

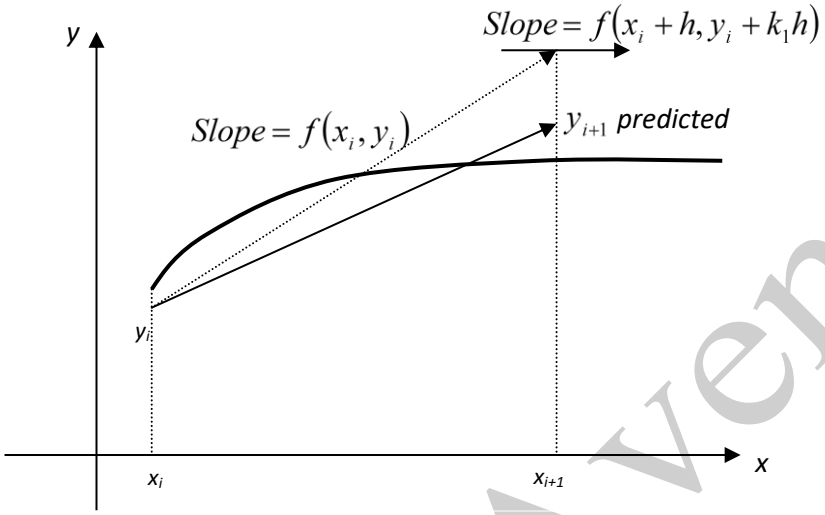


Figure 1 - Runge-Kutta 2nd order method (Heun's method).

Midpoint Method

Here $a_2 = 1$ is chosen, giving

$$a_1 = 0$$

$$p_1 = \frac{1}{2}$$

$$q_{11} = \frac{1}{2}$$

resulting in

$$y_{i+1} = y_i + k_2 h \quad (10)$$

where

$$k_1 = f\left(x_i, y_i\right) \quad (11a)$$

$$k_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1 h\right) \quad (11b)$$

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Ralston's Method

Here $a_2 = \frac{2}{3}$ is chosen, giving

$$a_1 = \frac{1}{3}$$

$$p_1 = \frac{3}{4}$$

$$q_{11} = \frac{3}{4}$$

resulting in

$$y_{i+1} = y_i + \left(\frac{1}{3}k_1 + \frac{2}{3}k_2\right)h \qquad (12)$$

where

$$k_1 = f\left(x_i, y_i\right) \text{ (13a)}$$

$$k_2 = f\left(x_i + \frac{3}{4}h, y_i + \frac{3}{4}k_1 h\right) \qquad (13b)$$

Question:

A ball at $1200\,\text{K}$ is allowed to cool down in air at an ambient temperature of $300\,\text{K}$. Assuming heat is lost only due to radiation, the differential equation for the temperature of the ball is given by

$$\frac{d\theta}{dt} = -2.2067\times10^{-12}\ (\theta^4 - 81\times10^8)$$

where θ is in K and t in seconds. Find the temperature at $t = 480$ seconds using Runge-Kutta 2nd order method. Assume a step size of $h = 240$ seconds.

Solution:

$$\frac{d\theta}{dt} = -2.2067\times10^{-12}\left(\theta^4 - 81\times10^8\right)$$

$$f\left(t,\theta\right) = -2.2067\times10^{-12}\left(\theta^4 - 81\times10^8\right)$$

As Per Heun's method given by Equations (8) and (9)

$$\theta_{i+1} = \theta_i + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right)h$$

$$k_1 = f\left(t_i, \theta_i\right)$$

$$k_2 = f\left(t_i + h, \theta_i + k_1 h\right)$$

$$i = 0, t_0 = 0, \theta_0 = \theta(0) = 1200$$

$$k_1 = f\left(t_0, \theta_o\right)$$

$$= f\left(0,1200\right)$$

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$$= -2.2067\times10^{-12}\left(1200^4 - 81\times10^8\right)$$
$$= -4.5579$$
$$k_2 = f\left(t_0 + h, \theta_0 + k_1 h\right)$$
$$= f\left(0 + 240, 1200 + \left(-4.5579\right)240\right)$$
$$= f\left(240, 106.09\right)$$
$$= -2.2067\times10^{-12}\left(106.09^4 - 81\times10^8\right)$$
$$= 0.017595$$
$$\theta_1 = \theta_0 + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right)h$$
$$= 1200 + \left(\frac{1}{2}\left(-4.5579\right) + \frac{1}{2}\left(0.017595\right)\right)240$$
$$= 1200 + \left(-2.2702\right)240$$
$$= 655.16\,\text{K}$$
$$i = 1, t_1 = t_0 + h = 0 + 240 = 240, \theta_1 = 655.16\text{K}$$
$$k_1 = f\left(t_1, \theta_1\right)$$
$$= f\left(240, 655.16\right)$$
$$= -2.2067\times10^{-12}\left(655.16^4 - 81\times10^8\right)$$
$$= -0.38869$$
$$k_2 = f\left(t_1 + h, \theta_1 + k_1 h\right)$$
$$= f\left(240 + 240, 655.16 + \left(-0.38869\right)240\right)$$
$$= f\left(480, 561.87\right)$$
$$= -2.2067\times10^{-12}\left(561.87^4 - 81\times10^8\right)$$
$$= -0.20206$$
$$\theta_2 = \theta_1 + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right)h$$
$$= 655.16 + \left(\frac{1}{2}\left(-0.38869\right) + \frac{1}{2}\left(-0.20206\right)\right)240$$
$$= 655.16 + \left(-0.29538\right)240$$
$$= 584.27\,\text{K}$$
$$\theta_2 = \theta\left(480\right) = 584.27\,\text{K}$$

The exact solution of the ordinary differential equation is given by the solution of a non-linear equation as

$$0.92593\ln\frac{\theta - 300}{\theta + 300} - 1.8519\tan^{-1}\left(0.0033333\theta\right) = -0.22067\times10^{-3}t - 2.9282$$

The solution to this nonlinear equation at $t = 480\,\text{s}$ is

$$\theta(480) = 647.57\,\text{K}$$

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5.4.3 Runge Kutta 4th order Method:

After reading this chapter, you should be able to

1. Develop Runge-Kutta 4th order method for solving ordinary differential equations,
2. Find the effect size of step size has on the solution,
3. Know the formulas for other versions of the Runge-Kutta 4th order method

Runge-Kutta 4th order method is a numerical technique used to solve ordinary differential equation of the form

$$\frac{dy}{dx} = f(x,y), y(0) = y_0$$

So only first order ordinary differential equations can be solved by using the Runge-Kutta 4th order method. In other sections, we have discussed how Euler and Runge-Kutta methods are used to solve higher order ordinary differential equations or coupled (simultaneous) differential equations.

The Runge-Kutta 4th order method is based on the following

$$y_{i+1} = y_i + (a_1k_1 + a_2k_2 + a_3k_3 + a_4k_4)h \quad (1)$$

where knowing the value of $y = y_i$ at x_i, we can find the value of $y = y_{i+1}$ at x_{i+1}, and

$$h = x_{i+1} - x_i$$

Equation (1) is equated to the first five terms of Taylor series

$$y_{i+1} = y_i + \frac{dy}{dx}\Big|_{x_i,y_i}(x_{i+1}-x_i) + \frac{1}{2!}\frac{d^2y}{dx^2}\Big|_{x_i,y_i}(x_{i+1}-x_i)^2 + \frac{1}{3!}\frac{d^3y}{dx^3}\Big|_{x_i,y_i}(x_{i+1}-x_i)^3 + \frac{1}{4!}\frac{d^4y}{dx^4}\Big|_{x_i,y_i}(x_{i+1}-x_i)^4 \quad (2)$$

Knowing that $\frac{dy}{dx} = f(x,y)$ and $x_{i+1} - x_i = h$

$$y_{i+1} = y_i + f(x_i,y_i)h + \frac{1}{2!}f'(x_i,y_i)h^2 + \frac{1}{3!}f''(x_i,y_i)h^3 + \frac{1}{4!}f'''(x_i,y_i)h^4 \quad (3)$$

Based on equating Equation (2) and Equation (3), one of the popular solutions used is

$$y_{i+1} = y_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)h \quad (4)$$

$$k_1 = f(x_i,y_i) \quad (5a)$$

$$k_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1h\right) \quad (5b)$$

$$k_3 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_2h\right) \quad (5c)$$

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$$k_4 = f\left(x_i + h, y_i + k_3 h\right) \quad (5d)$$

Range- Kutte Method:-

Let $\frac{dy}{dx} = f\left(x, y\right)$ and $y = y_0 \ at \ x = x_0 = a$ or $y\left(x_0\right) = value of \ y \ is \ y_0 \ at \ x = x_0$

The formula for the solution of $\frac{dy}{dx} = f\left(x, y\right)$ by Range kutte method at $x = a + h$ is $y_1 = y_0 + k$

a) 2nd Order Range- Kutte Method:-

$$y_1 = y_0 + k = y_0 + \frac{1}{2}\left(k_1 + k_2\right)$$

Where $k = \frac{1}{2}\left(k_1 + k_2\right)$

and $k_1 = h\, f\left(x_0, y_0\right),\ k_2 = h\, f\left(x_0 + h, y_0 + k_1\right)$

b) 4th Order Range- Kutte Method:

$$y_1 = y_0 + k \ = y_0 + \frac{1}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right)$$

Where $k = \frac{1}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right)$

and $k_1 = h\, f\left(x_0, y_0\right)$, $k_2 = h\, f\left(x_0 + \frac{h}{2},\ y_0 + \frac{k_1}{2}\right)$

$$k_3 = h\, f\left(x_0 + \frac{h}{2},\ y_0 + \frac{k_2}{2}\right)$$

$$k_4 = h\, f\left(x_0 + h, y_0 + k_3\right)$$

Points to remember:

1. R - K methods do not require prior computations of the higher derivatives of y(x) as Taylor series does.
2. The R – K formulae involve the computation of f(x,y) at various positions and this function occurs in the given equation.
3. To evaluate y_{i+1} we need information only at y_i. Information at y_{i-1}, y_{i-2} etc are not directly required hence
   R – K methods are one step methods.
4. These methods agree with Taylor series solution upto the terms of $h^{\wedge r}$, where 'r' differs from method to method and is known as order of R – K method.

Question:

A ball at 1200 K is allowed to cool down in air at an ambient temperature of 300 K. Assuming heat is lost only due to radiation, the differential equation for the temperature of the ball is given by

$$\frac{d\theta}{dt} = -2.2067 \times 10^{-12}\left(\theta^4 - 81 \times 10^8\right), \theta\left(0\right) = 1200\,\mathrm{K}$$

where θ is in K and t in seconds. Find the temperature at $t = 480$ seconds using Runge-Kutta 4th order method. Assume a step size of $h = 240$ seconds.

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Solution:

$$\frac{d\theta}{dt} = -2.2067\times10^{-12}\left(\theta^4 - 81\times10^8\right)$$

$$f(t,\theta) = -2.2067\times10^{-12}\left(\theta^4 - 81\times10^8\right)$$

$$\theta_{i+1} = \theta_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)h$$

For $i = 0$, $t_0 = 0$, $\theta_0 = 1200\text{K}$

$$k_1 = f(t_0, \theta_0)$$
$$= f(0, 1200)$$
$$= -2.2067\times10^{-12}\left(1200^4 - 81\times10^8\right)$$
$$= -4.5579$$

$$k_2 = f\left(t_0 + \frac{1}{2}h, \theta_0 + \frac{1}{2}k_1 h\right)$$
$$= f\left(0 + \frac{1}{2}(240), 1200 + \frac{1}{2}(-4.5579)\times240\right)$$
$$= f(120, 653.05)$$
$$= -2.2067\times10^{-12}\left(653.05^4 - 81\times10^8\right)$$
$$= -0.38347$$

$$k_3 = f\left(t_0 + \frac{1}{2}h, \theta_0 + \frac{1}{2}k_2 h\right)$$
$$= f\left(0 + \frac{1}{2}(240), 1200 + \frac{1}{2}(-0.38347)\times240\right)$$
$$= f(120, 1154.0)$$
$$= -2.2067\times10^{-12}\left(1154.0^4 - 81\times10^8\right)$$
$$= -3.8954$$

$$k_4 = f(t_0 + h, \theta_0 + k_3 h)$$
$$= f(0 + 240, 1200 + (-3.894)\times240)$$
$$= f(240, 265.10)$$
$$= -2.2067\times10^{-12}\left(265.10^4 - 81\times10^8\right)$$
$$= 0.0069750$$

$$\theta_1 = \theta_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)h$$
$$= 1200 + \frac{1}{6}\left(-4.5579 + 2(-0.38347) + 2(-3.8954) + (0.069750)\right)240$$
$$= 1200 + (-2.1848)\times240$$
$$= 675.65\,\text{K}$$

θ_1 is the approximate temperature at,

$t = t_1$

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$= t_0 + h$

$= 0 + 240$

$= 240$

$\theta_1 = \theta(240)$

$\approx 675.65\,\text{K}$

For $i = 1, t_1 = 240, \theta_1 = 675.65\,\text{K}$

$k_1 = f(t_1, \theta_1)$

$= f(240, 675.65)$

$= -2.2067 \times 10^{-12}\left(675.65^4 - 81 \times 10^8\right)$

$= -0.44199$

$k_2 = f\left(t_1 + \frac{1}{2}h, \theta_1 + \frac{1}{2}k_1 h\right)$

$= f\left(240 + \frac{1}{2}(240), 675.65 + \frac{1}{2}(-0.44199)240\right)$

$= f(360, 622.61)$

$= -2.2067 \times 10^{-12}\left(622.61^4 - 81 \times 10^8\right)$

$= -0.31372$

$k_3 = f\left(t_1 + \frac{1}{2}h, \theta_1 + \frac{1}{2}k_2 h\right)$

$= f\left(240 + \frac{1}{2}(240), 675.65 + \frac{1}{2}(-0.31372) \times 240\right)$

$= f(360, 638.00)$

$= -2.2067 \times 10^{-12}\left(638.00^4 - 81 \times 10^8\right)$

$= -0.34775$

$k_4 = f(t_1 + h, \theta_1 + k_3 h)$

$= f(240 + 240, 675.65 + (-0.34775) \times 240)$

$= f(480, 592.19)$

$= 2.2067 \times 10^{-12}\left(592.19^4 - 81 \times 10^8\right)$

$= -0.25351$

$\theta_2 = \theta_1 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)h$

$= 675.65 + \frac{1}{6}\left(-0.44199 + 2(-0.31372) + 2(-0.34775) + (-0.25351)\right) \times 240$

$= 675.65 + \frac{1}{6}(-2.0184) \times 240$

$= 594.91\text{K}$

θ_2 is the approximate temperature at,

$t = t_2$

$= t_1 + h$

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$= 240 + 240$
$= 480$
$\theta_2 = \theta(480)$
$\approx 594.91\text{K}$

5.4.4. Picard's Iterative Method:

Let $\frac{dy}{dx} = f(x, y)$ and $y = y_0$ *at* $x = x_0$

The formula for the solution of $\frac{dy}{dx} = f(x, y)$ by Picard's Iterative Method for n[th] iteration is

$$y^{(n)} = y_0 + \int_{x_0}^{x} f\left(x, y^{(n-1)}\right) dx$$

$$y^{(1)} = y_0 + \int_{x_0}^{x} f\left(x, y_0\right) dx$$

$$y^{(2)} = y_0 + \int_{x_0}^{x} f\left(x, y^{(1)}\right) dx$$

……

……

and so on.

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SET- 12
Solution of Non-Algebraic Equation

1. For the *equation* $f(x)=0\Rightarrow x^3-4x-9=0$, the roots by using the bisection method in four stages is.

   a) 2.7065 b) 2.6875 c) 2.750 d) None of these

2. For the *equation* $f(x)=0\Rightarrow 2x-\log_{10}x-7=0$, the roots by using Regular- Falsi method correct to three places of decimal is

   a) 3.683 b) 3.789 c) 3.788 d) 3.790

3. For the series $x_{n+1}=\frac{x_n}{2}+\frac{9}{8x_n}$, $x_n=0.5$ *obtained by Newtion Rapson Method*, the series converges to

   a) 1.5 b) $\sqrt{2}$ c) 1.6 d) 1.4

4. For $a>0$, If we have to calculate its reciprocal value $1/a$ by using the Newton Raphson Method for $f(x)=0$ .The Algorithm for the function by Newton Raphson Method will be

   a) $x_{n+1}=\frac{1}{2}\left(x_n+\frac{a}{x_n}\right)$ b) $x_{n+1}=\left(x_n+\frac{a}{2}x_n^{\,2}\right)$

   c) $x_{n+1}=\left(2x_n-a\,x_n^{\,2}\right)$ d) $x_{n+1}=\left(x_n+\frac{a}{2}x_n^{\,2}\right)$

5. For $a>0$, If we have to calculate its reciprocal value $1/a$ by using the newton Raphson Method for $f(x)=0$. For $a=7$ starting with $x=0.2$, the first two iterations will be

   a) 0.11 ,0.1299 b) 0.12 ,0.1392

   c) 0.12 ,0.1416 d) 0.13 ,0.1428

6. For finding the square root of 2, the Newton Raphson Algorithm will be

   a) $x_{n+1}=\frac{1}{2}\left(x_n+\frac{2}{x_n}\right)$ b) $x_{n+1}=\frac{1}{2}\left(x_n-\frac{2}{x_n}\right)$

   c) $x_{n+1}=\left(\frac{2}{x_n}\right)$ d) $x_{n+1}=\sqrt{2+x_n}$

7. For equation $x^2=3$, the root after second iteration of Newton Raphson Method by taking $x_0=3$ is,

   a) $347/200$ b) $97/56$ c) $7/4$ d) $7/2$

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8. For the equation $x^3 - 5x + 3 = 0$, the value of x_0 *to get the solution in the interval* $(0.5, 0.75)$ s

a) 0.5 b) 0.625 c) 0.75 d) *None of these*

9. For the equation $e^x + x - 3 = 0$ the roots lie in the interval

a) $(0,1)$ b) $(1,2)$ c) $(2,3)$ d) $(2,4)$

10. For the equation $x - e^x = 0$ the roots lie in the interval

a) $(0,1)$ b) $(-1,0)$ c) $(1,2)$ d) $(2,3)$

11. For the equation $x^2 - 9x + 1 = 0$ the root by using Regula-Falsi Method after first iteration is

a) 2.47 . b) 2.5 c) 3.57 d) 3

12. For the equation $x^3 - x - 5 = 0$, the root lies between $1 \textit{and } 2$ after first iteration by Newton- Raphson method is

a) 3.546 b) 3.500 c) 1.904 d) 1.909

13. The roots of the equation $x^3 - 6x - 1 = 0$ lies in the interval

a) $(4,6)$ b) $(3,5)$ c) $(3,4)$ d) $(2,3)$

14. For the eqaution $x^3 - 3x - 5 = 0$, by using Newton-Raphson Method the root lies between $2 \textit{and } 2.5$ after first iteration is

a) 2.45 b) 2.35 c) 2.33 d) 2.25

15. For the equation $x^3 + 4x - 9 = 0$, by using the Newton - Raphson Method, The iterative equation will be

a) $x_{n+1} = \dfrac{2x_n^{\,3} + 9}{3x_n^{\,2} + 4}$ b) $x_{n+1} = \dfrac{3x_n^{\,2} + 4}{2x_n^{\,2} + 9}$

c) $x_{n+1} = \dfrac{4x_n^{\,2} + 3}{9x_n^{\,2} + 2}$ d) $x_{n+1} = x_n - 3x_n^{\,2} + 4$

16. For the equation $x^2 + 3x - 7 = 0$, by using the Newton- Raphson Method with $x_0 = 1$, the value of x_1 will be

a) 0.5 b) 1.406 c) 1.5 d) 2

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Solutions:

1. Ans: b)

Let $f(c) = x^3 - 4x - 9$
We get $f(2) = -9$ and $f(3) = 6$
Since f(2) is negative and f(3) is positive, a root lies between 2 and 3 first approximation"

$$x_1 = \frac{1}{2}(2+3) = 25$$

Then $f(x_1) = 25^3 - 4(25) - 9 = -3375$ (-ve), root lies between x_1 and 3
Second approximation:

$$x_2 = \frac{1}{2}(x_1 + 3) = 2.75$$

Then $f(x_2) = (2.75)^3 - 4(2.75) - 9 = 0.7969$ (+ve), root lies between x_1 and x_2
Third approximation

$$x_3 = \frac{1}{2}(x_1 + x_2) = 2625$$

Then $f(x_3) = (2.625)^3 - 4(2.625) - 9 = -1.4121$(-ve). Root lies between x_2 and x_3
Fourth approximation:

$$x_4 = \frac{1}{2}(x_2 + x_3) = 2.6875.$$

Hence the root is 2.6875 approximately
Hence B is correct option

2. Ans: b)

We have $f(x) = 2x - \log_{10} x - 7$

We get f(35) = - 0.5441 and f(4) = 0.3979, so a root lies between 3.5 and 4

Taking $x_0 = 35$, x1 = 4, in the method of false position, we get

$$x_2 = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_0)$$

$$= 35 - \frac{0.5}{0.3979 + 0.5441}(-0.5441) = 3.7888$$

Since f(3.7888) = - 0.009 and f(4) = 0.3979, therefore the root lies between 3.7888 and 4 taking x_0 = 3.7888, x_1 = 4, we obtain

$$x_3 = 3.7888 - \frac{0.2112}{0.3988}(-0.009) = 3.7893$$

Thus the required root correct to three places of decimal is 3.789

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3. Ans: a) for the series to be convergent 2 consecutive roots must be same i.e $x_{n+1} = x_n$

We have $x_{n+1} = \frac{x_x}{2} + \frac{9}{8x_n}$

Now, $x_1 = \frac{x_0}{2} + \frac{9}{8x_0}$ where x_0 = 0.5

$= \frac{0.5}{2} + \frac{9}{8(0.5)} = 25$

$x_2 = 1.7$

$x_3 = 1.5117$

$x_4 = 1.5000$

$x_5 = 1.500$

Hence A is correct option

4. Ans:c)

We have $x = \frac{1}{a}$ *or* $xa - 1 = 0$

Or f(x) = xa -1

Using Newton – Raphson algorithm, $x_{k+1} = x_k - \frac{f(x_k)}{f(x_k)}$

$= x_k - \frac{x_k\, a - 1}{a}$

$= x_4 - \frac{x_4\, a - 1}{\frac{1}{x_k}}$ $x = \frac{1}{a} \rightarrow a = \frac{1}{x}$

$x_{k+1} = 2x_k - ax_k^2$

Hence C is correct option

5. Ans: b)

For a = 7and x_0 = 0.2 we have

$x_1 = 02(02) - 7(0.2)^2 = 0.12$

Taking x_1 = 0.12 $x_2 = 0.2(0.12) - 7(0.12)^2 = 0.1392$

Hence B is Correct option

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6. Ans: a)

We have $x=\sqrt{2}$ *or* $x^2-2=0$

Or $f(x)=x^2-2$

From Newton- Raphson iteration

$$x_{n+1}=x_m-\frac{f(x_n)}{f(x_n)}$$

$$=x_n-\frac{x_n^2-2}{2x_n}$$

$$=\frac{1}{2}\left(x_n+\frac{2}{x_n}\right)$$

Hence A is correct option

7. Ans: b)

We have $f(x) = x^2 - 3$

$F(x) = 2x$

We get $f\left(x_0=\frac{3}{2}\right)=-\frac{3}{4}$ *and* $f'\left(x_0=\frac{3}{2}\right)=3$

First iteration by Newton-Raphson method is

$$x_1=x_0-\frac{f(x_0)}{f'(x_0)}$$

$$=\frac{3}{2}-\frac{-\frac{3}{4}}{3}=\frac{7}{4}$$

Second iteration $x_2=\frac{7}{4}-\frac{\frac{1}{16}}{\frac{7}{2}}=\frac{97}{56}$

Hence B is correct option

8. Ans: c)

We have $f(x) = x^3 - 5x+3$

$f(0.5) = (0.5)^3 - 5\ (0.5) + 3 = 0.625$

$f(0.75) = (0.75)^3 - 5\ (0.75) + 3 = -\ 0.328$

Since $|f(0.5)|<|f(0.75)|$, thus value of x_0 will be 0.75

Hence C is correct option.

9. Ans: a)

$f(0) = e^0 + 0.3 = -2 < 0$

$f(1) = e^1 + 1\text{-}3 = 0.726 > 0$

$f(2) = e^3 + 2\ \text{-}3 = 6.39 > 0$

$f(3) = e^3 + 3 - 3 = 20.090 > 0$

$f(4) = e^4 + 4 - 3 = 55.60 > 0$

Thus root liw in (0,1)

Hence A is correct option

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10. Ans: a)

$f(x) = x-e^{-x}$
$f(-1) = -1 -e^{-(-1)} = -3.71$
$f(0) = 0 – e^{-0} = -1$
$f(1) = 1 – e^{-1} = 0.63$
$f(2) = 2 – e^{-2} = 1.86$
$f(3) = 3 – e^{-3} = 2.95$

It can be easily seen that root lie in (0,1)
Hence A is correct option

11. Ans: a)

Hare $x_0 = 2$ and $x_1 = 4$; $f(x_0) = f(2) = -9$ and $f(x_1) = f(4) = 29$
By Regula-falsi method we know

$$x_2 = x_0 - \frac{(x_1 - x_0) f(x)}{f(x_1) - f(x_0)}$$

$$= 2 - \frac{(4-2)(-9)}{29-(-9)}$$

$$= 2 + \frac{18}{38} = 247$$

Hence A is correct option

12. Ans: d)

We have $f(x) = x^3 - x - 5$

Or $f'(x) = 3x^2 - 1$

We get f(1) = 1-1-5=-5 and f(2) = 8-2-5 = 1

This indicate that the root in nearer to 2 than 1. Let initial value x_0 of the root be 2

$$f'(x_0) = f'(2) = 3(2)^2 - 1 = 11$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 2 - \frac{1}{11} = 1.90909$$

Career Avenues

13. Ans: d)

We Have $f(x) = x^3 - 6x - 1$

$f(2) = 2^3 - 6\,(2)^{-1} = -5$

$f(3) = 3^3 - 6\,(3)^{-1} = 8$

$f(4) = 4^3 - 6\,(4) - 1 = 39$

Since f(2) < 0 and f(3) > 0, thus root lie between (2,3) Hence D is correct option

14. Ans: c)

We have $f(x) = x^3 - 3x - 5$

Or $f(x) = 3x^2 - 3$

Now $f(2) = -3$ and $f(25) = 4.625$

We have |f(2)| > |f(25)|, this indicate that root is nearer to 2 than 2.5

Let initial value of x_0 of the root be 2

$f'(2) = 3(2)^2 - 3 = 9$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 2 - \frac{-3}{9} = 2.33$$

Hence C is correct option

15. Ans: a)

We have $f(x) = x^3 + 4x - 9 = 0$

According to Newton-Raphson's method

$$x_{k+1} = x_4 - \frac{f(x_k)}{f'(x_k)}$$

Or $$x_{k+1} = x_4 - \frac{x_k^3 + 4x_k - 9}{3x_k^2 + 4}$$

Or $$x_{k+1} = \frac{3x_k^3 + 4x_k - x_k^3 - 4x_k + 9}{3x_k^2 + 4}$$

Or $$x_{k+1} = \frac{2x_k^2 + 9}{3k_k^2 + 4}$$

Hence A is correct option

Career Avenues

16. Ans: c)

We have $f(x) = x^3 + 3x - 7$

$f(x) = 3x^2 + 3$

and $f(1) = -3$ and $f'(1) = 6$

From Newton-Raphson method we know that

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Taking $x_0 = 1$

$$x_1 = 1 - \left(\frac{-3}{6}\right) = 1 - (-0.5) = 1.5$$

Hence C is correct option

Career Avenues

SET- 13
Numerical Integration

1. For $f(x)=x^4$, By using the Simpson's rule the value of $\int_{-3}^{3} x^4\,dx$ for 6 sub-intervals will be
   a) 96
   b) 98
   c) 99
   d) 100

2. For $f(x)=\frac{1}{x}$, By using the Simpson's rule the value of $\int_{1}^{2} \frac{1}{x}dx$ *by dividing* the interval $(1,2)$ into 4 sub-intervals is
   a) 0.6932
   b) 0.6753
   c) 0.6692
   d) 0.6319

3. For $f(x)=\sqrt{1-0.162\sin^2 x}\,dx,$ by taking step size $\frac{\pi}{12}$, the value of $\int_{0}^{\pi/2} f(x)\,dx$, by Simpson's $\frac{1}{3}rd$ rule is
   a) 1.5058
   b) 1.5759
   c) 2.5056
   d) 1.5056

4. For the function $f(x)=x^3$, the value of $\int_{6}^{1} f(x)\,dx$, by using Trapezoidal rule taking five sub intervalsis
   a) 0.21
   b) 0.23
   c) 0.24
   d) 0.26

5. For the function $f(x)=\frac{1}{1+x}$, the value of $\int_{0}^{1} f(x)\,dx$ by taking four sub-intervals in Simpson's method is
   a) 0.6035
   b) 0.6945
   c) 0.6145
   d) 0.5945

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6. For $f(x)=\frac{1}{1+x^2}$, By Simpson's rule, If $\int_0^1 f(x)dx = \frac{1}{12}\left[3.1+4(a+b)\right]$ for the interval $[0,1]$ divided in four sub-intervals and a *and* b are the values of $f(x)$ at two of its division points, then the value of a *and* b *respectively is*.

a) $\frac{1}{1.0625}, \frac{1}{1.25}$

b) $\frac{1}{1.0625}, \frac{1}{1.5625}$

c) $\frac{1}{1.25}, 1$

d) $\frac{1}{1.5625}, \frac{1}{1.25}$

7. For $f(x)= e^x$, If $e^0=1,\ e^1=2.72,\ e^2=7.39,\ e^3=20.09$ and $e^4=54.60$, then by Simpson's rule $\int_0^4 f(x)dx$ *is*

a) 5.387
b) 52.78
c) 53.18
d) 53.87

Career Avenues

Solutions:

1. Ans: b)

Using Simpson rule, we have

$$\int_{-3}^{3} x^4\,dx = \frac{h}{3}\left[y_0 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4) + y_6\right]$$

$$= \frac{1}{3}\left[81 + 4(16 + 0 + 16) + 2(1 + 1) + 81\right]$$

$$= \frac{1}{3}\left[81 + 128 + 4 + 81\right]$$

$$= 98$$

Hence B is correct option

2. Ans: a)

$$h = \frac{2-1}{4} = 0.25$$

Using Simpson rule we have

$$\int_{1}^{2} \frac{dx}{x} = \frac{h}{3}\left[y_0 + 4(y_1 + y_3) + 2y_2 + y_4\right]$$

$$= \frac{0.25}{3}\left[1 + 4(0.8 + 0.571) + 2(0.667) + 0.5\right]$$

$$= 0.6932$$

Hence A is correct option

3. Ans: d)

$$f(x) = \sqrt{1 - 0.162\sin^2 x}$$

$$= \sqrt{1 - 0.162\frac{(1 - \cos 2x)}{2}}$$

$$= \sqrt{0.919_0.081\cos 2x}$$

Here where $h = \pi/12$

When $x_0 = 0$, $y_0 = 1$

When $x_1 = \pi/12$, $y_1 = \sqrt{0.919 + 0.081\cos 30^O} = 0.9795$

When $x_2 = \pi/6$, $y_2 = \sqrt{0.919 + 0.081\cos 60^O} = 0.9795$

When $x_3 = \pi/4$, $y_3 = \sqrt{0.919 + 0.081\cos 90^O} = 0.9586$

When $x_4 = \pi/3$, $y_4 = 0.9372$

When $x_5 - 5\pi/12$, $y_5 = 0.9213$

When $x_6 = \pi/2$, $y_6 = 0.9154$

Career Avenues

$$\int_0^{\pi/2} f(x)\,dx = \frac{h}{2}\left[(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)\right]$$

= 1.5056

Hence D is correct option

4. Ans: d)

$$\int_0^1 x^3\,dx = \frac{h}{2}\left[y_0 + (y_1 + y_2 + y_3 + y_4) + y_5\right]$$

$$= \frac{0.2}{2}\left[0 + 2(0.008 + 0.064 + 0.216 + 0.512) + 1\right]$$

5. Ans: b)

Using Simpson's formula

$$\int_0^1 \frac{1}{1+x}\,dx = \frac{h}{3}\left[y_0 + 4(y_1 + y_3) + 2y_2 + y_4\right]$$

$$= \frac{0.25}{3}\left[1 + 4(0.8 + 0.571) + 2(0.67) + 0.5\right]$$

=0.6945

Hence B is correct option

6. Ans: b)

$$a = y_1 = \frac{1}{1+x^2} = \frac{1}{1+(0.25)^2} = \frac{1}{1.0625}$$

$$b = y_3 = \frac{1}{1+x_3^2} = \frac{1}{1+(0.75)^2} = \frac{1}{1.5625}$$

Hence B is correct option

7. Ans: d)

Here h=1

By Simpson rule we have

$$\int_0^4 e^x\,dx = \frac{1}{3}\left[1 + 4(2.72 + 20.09) + 2(7.39) + 54.60\right]$$

$$= \frac{1}{3}\left[1 + 91.24 + 14.78 + 54.60\right]$$

= 53.87

Hence D is correct option

Career Avenues

SET- 14
Solution of Differential Equation by Numerical Methods

1. For $\frac{dy}{dx} = 1 + xy = f(x, y)$, it is given that $y = 1 at\ x = 0$. By using Taylor's series mehod, The value of $y(0.1)$ correct to four decimal places is $(h = 0.1)$

   a) 2.1513
   b) 1.1053
   c) 1.2689
   d) *None of the above*

2. The differential equation $\frac{dy}{dx} = x - y^2\ is\ solved$ by using Taylor's series method, the value of $y(0.1)$ correct upto four decimal places is $(given\ that\ y = 1 at\ x = 0)$

   a) 1.4396
   b) 0.9138
   c) 1.0134
   d) 0.9159

3. The differential eqaution $\frac{dy}{dx} = f(x)$, $y(0) = 0$ is solved by 4th order Runge- Kutta method with step size h. The solution at $x = h$ is given by

   a) $y(h) = \frac{h}{6}\left[f(0) + 4 f\left(\frac{h}{2}\right) + f(h) \right]$
   b) $y(h) = \frac{h}{6}\left[f(0) + 2 f\left(\frac{h}{2}\right) + f(h) \right]$
   c) $y(h) = \frac{h}{6}\left[f(0) + f\left(\frac{h}{2}\right) + f(h) \right]$
   d) $y(h) = \frac{h}{6}\left[2 f(0) + 4 f\left(\frac{h}{2}\right) + f(h) \right]$

4. The differential eqaution $\frac{dy}{dx} = y$, $y(0) = y_0$. is solved by 2nd order Runge- Kutta method with step size h. The solution at $x = h$ is given by

   a) $y_0 (h - 1)^2$
   b) $\frac{y_0}{2}\left(h^2 - 2h + 2\right)$
   c) $\frac{y_0}{6}\left(h^2 - 2h + 2\right)$
   d) $y_0\left(1 + h + \frac{h^2}{2} + \frac{h^2}{6}\right)$

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5. For $\frac{dy}{dx} = x + y^2$, $given that\ y = 1 at\ x = 0.$ The differential eqaution is solved by 4th order Runge- Kutte method, then the value of $y\, at\, x = 0.2\, is\, (h = 0.2)$.

a) 1.2735 b) 2.1635 c) 1.9356 d) 2.9468

6. The differential eqaution $\frac{dy}{dx} = x + y^2$, $with\ y = 0\, at\, x = 0, is\, solved\, by\, using$ Picard's Method up to 3rd order of approximation, then the solution is

a) $\frac{x^2}{2} + \frac{x^5}{40} + \frac{x^8}{480} + \frac{x^{11}}{1600}$ b) $\frac{x^2}{2} + \frac{x^5}{20} + \frac{x^8}{160} + \frac{x^{11}}{4400}$

c) $\frac{x^2}{2} + \frac{x^5}{20} + \frac{x^8}{160} + \frac{x^{11}}{2400}$ d) $\frac{x^2}{2} + \frac{x^5}{40} + \frac{x^8}{480} + \frac{x^{11}}{2400}$

7. For $\frac{dy}{dx} = y - x$ $given that\ y = 2 at\ x = 0.$ Using Picard's method up t 3rd order of approximation the solution of the eqaution is

a) $2 + 2x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{24}$ b) $2 + 2x + \frac{x^2}{2} + \frac{x^3}{6}$

c) $2 + 2x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{16}$ d) $2 + 2x + \frac{x^2}{2} - \frac{x^3}{6}$

Career Avenues

Solutions:

1. Ans: b)

We have h = 0.1, $x_0 = 0$, $x_1 = x_0 + h = 0.1$

Now $\frac{dy}{dx} = 1 + xy$

Or $\frac{d^2y}{dx^2} = x\frac{dy}{dx} + y$

Or $\frac{d^3y}{dx^3} = x\frac{d^2y}{dx^2} + 2\frac{dy}{dx}$

Or $\frac{d^4y}{dx^4} = x\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2}$

From y (0) =1 we have $\frac{dy}{dx} = 1; \frac{d^2y}{dx^2} = 1, \frac{d^3y}{dx^3} = 2, \frac{d^4y}{dx^4} = 3$ and so on

Taylor series expression given

$$y(x+h) = y(x) + h\frac{dy}{dx} + \frac{h^2}{2!}\frac{d^2y}{dx^2} + \frac{h^3}{3!}\frac{d^3y}{dx^3} + \frac{h^4}{4!}\frac{d^4y}{dx^4} + \ldots$$

$$y(0.1) = 1 + 0.1 \times 1 + \frac{(0.1)^2}{2!}1 + \frac{(0.1)^3}{3!}2 + \ldots$$

$$= 1 + 0.1 + \frac{0.01}{2} + \frac{0.001}{3} + \ldots$$

$$= 1 + 0.1 + 0.005 + 0.000033\ldots.$$

$$= 1.1053$$

2. Ans: b)

Let h =0.1, Given $x_0 = 0, y_0 = 1 \quad x_1 = x_0 + h = 0.1,$

Now $\frac{dy}{dx} = x - y^2 \quad and \left(\frac{dy}{dx}\right)_{0,1} = -1$

$\frac{d^2y}{dx^2} = 1 - 2y\frac{dy}{dx} \quad and \left(\frac{d^2y}{dx^2}\right)_{0,1} = 3$

$\frac{d^3y}{dx^3} = -2\left(\frac{dy}{dx}\right)^2 - 2y\frac{d^2y}{dx^2} \quad and \left(\frac{d^2y}{dx^3}\right)_{0.1} = -3$

$\frac{d^4y}{dx^4} = -2\left[3\frac{dy}{dx}\frac{d^2y}{dx^2} + y\frac{d^2y}{dx^3}\right] \quad and \left(\frac{d^4y}{dx^4}\right)_{0.1} = 34$

The Taylor series expression gives

$$y(x+h) = y(x) + h\frac{dy}{dx} + \frac{h^2}{2!}\frac{d^2y}{dx^2} + \frac{h^3}{3!}\frac{d^3y}{dx^3} + \frac{h^4}{4!}\frac{d^4y}{dx^4} + \ldots$$

$$y(0.1) = 1 + 0.1(-1) + \frac{(0.1)^2}{2!} + \frac{(0.1)^3}{3!}(-8) + \frac{(0.1)^4}{4!}34 + \ldots.$$

$$= 1 - 0.1 + 0.0015 - 0.001333 + 0.0001417 = 0.9138$$

Hence B is Correct option

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3. Ans: a)

We have $\frac{dy}{dx} = f(x)$

Step size = $h, y_0 = 0, x_0 = 0$

By runge-kutta of the fourth order

$$k_1 = hj(x_0, y_0) = h\left[f(x_0)\right] = hf(0)$$

$$k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right) = hf\left(x_0 = \frac{1}{2}h\right) = hf\left(\frac{h}{2}\right)$$

$$k_3 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right) = hf\left(\frac{h}{2}\right)$$

$$k_3 = hf(x_0 + h, y_0 + k_3) = hf(h)$$

Hance by fourth order runge – kutta formula the solution at x = h

$$y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$y(h) = 0 + \frac{1}{6}\left[hf(0) + ahf\left(\frac{h}{2}\right) + ahf\left(\frac{h}{2}\right) + hf(h)\right]$$

$$= \frac{h}{6}\left[f(0)4f\left(\frac{h}{2}\right) + f(h)\right]$$

Hance A is correct option

4. Ans: b)

$$k_2 = hf(x_0 + h, k_1)$$

$$\frac{dy}{dx} = y = f(x, y)$$

$$y(0) = y_0$$

$$k_1 = -hy_0$$

$$k_2 = hf(h, y_0 + k_1)$$

$$= h\left[-1(y_0 + k_1)\right]$$

$$= h\left[-y_0 + hy_0\right]$$

$$= -hy_0(1 - h)$$

$$k = \frac{1}{2}\left[(-hy_0) + \left\{-hy_0(1-h)\right\}\right]$$

$$= \frac{1}{2}\left[-2hy_0 + h^2 y_0\right]$$

$$= \frac{h^2}{2}y_0 - hy_0$$

$$y_1 = y_0 + \frac{h^2}{2}y_0 - hy_0$$

$$= \frac{y_0}{2}\left[h^2 - 2h + 2\right]$$

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5. Ans: a)

To find $y_1 = y_{(0.2)}$

$$k_1 = hf(x_0, y_0) = (0.2) f(0,1) = (0.2) \times 1 = 0.2$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$= (0.2) f(0.1, 1.1) = 0.2(1.31) = 0.262$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$$

$$= 0.2 f(0.1, 1.131) = 0.2758$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

$$= (0.2) f(0.2, 1.2758) = 0.3655$$

$$k = \frac{1}{6}[k_1 + 2k_2 + 2k_3 + 2k_4]$$

$$= \frac{1}{6}[0.2 + 2(0.262) + 2(0.2758) + 0.3655] = 0.2735$$

Here $y_1 = y_{(0.2)} = y_0 + k = 1 + 0.2735 \Rightarrow 1.2735$

Hence A is correct option

6. Ans: a)

Here f(x,y) = x + y^2, $x_0 = 0$ $y_0 = 0$

We have, by Picard's method

$$y = y_0 + \int_{x_0}^{x} f(x, y_0)\, dx$$

The first approximation by y is given by

$$y^{(1)} = y_0 + \int_{x_0}^{x} f(x, y_0)\, dx = 0 + \int_{0}^{x} f(x, 0) dx$$

$$= 0 + \int_{0}^{x} x dx = \frac{x^2}{2}$$

The second approxmation to y is given by

$$y^{(2)} = y_0 + \int_{x_0}^{x} f\left(x, y^{(1)}\right) dx = 0 + \int_{0}^{x} f\left(x, \frac{?}{2}\right) dx$$

$$= \int_{0}^{x} \left(x + \frac{x^1}{4}\right) dx = \frac{x^2}{2} + \frac{x^5}{50}$$

The second approximation to y is given by

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$$y^{(2)} = y_0 + \int_{x_0}^{x} f\left(x, y^{(1)}\right) dx$$

$$= 2 + \int_{x_0}^{x} f\left(x, 2 + 2x - \frac{x^2}{2}\right) dx$$

$$= 2 + \int_{0}^{2} \left(2 + 2x - \frac{x^2}{2} - x\right) dx$$

$$= 2 + 2x + \frac{x^2}{2} - \frac{x^3}{6} \qquad ..(2)$$

The third approxmation to y is given by

$$y^{(3)} = y_0 + \int_{x_0}^{x} f\left(x, y^{(2)}\right) dx$$

$$= 2 + \int_{x_0}^{x} f\left(x, 2 + 2x + \frac{x^2}{2} - \frac{x^3}{6}\right) dx$$

$$= 2 + \int_{0}^{x} \left(2 + 2x + \frac{x^2}{2} - \frac{x^3}{6} - \right) dx$$

$$= 2 + 2x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{24}$$

Hence A is Correct option

7. Ans:a)

Here f (x,y) = y – x; $x_0 = 0$, $y_0 = 2$

We have by Picard method

$$y = y_0 + \int_{x_0}^{x} f(x, y)\, dx$$

The first approxmation to y is given by

$$y^{(1)} = y_0 + \int_{x_0}^{x} f(x, y_0) dx = 2 + \int_{0}^{x} f(x, 2) dx$$

$$= 2 + \int_{0}^{x} (2 - x)\, dx = 2 + 2x - \frac{x^2}{2}$$

The second approximation to y given by

$$y^{(2)} = y_0 + \int_{x_0}^{x} f\left(x, y^{(1)}\right) dx$$

$$= 2 + \int_{x_0}^{x} f\left(x, 2 + 2x - \frac{x^2}{2}\right) dx$$

$$= 2 + \int_{0}^{2} \left(2 + 2x - \frac{x^2}{2} - x\right) dx$$

$$= 2 + 2x + \frac{x^2}{2} - \frac{x^3}{6} \qquad ...(2)$$

The third approxmation to y is given by

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$$y^{(3)} = y_0 + \int_{x_0}^{x} f\left(x, y^{(2)}\right) dx$$

$$= 2 + \int_{x_0}^{x} f\left(x, 2 + 2x + \frac{x^2}{2} - \frac{x^3}{6}\right) dx$$

$$= 2 + \int_{0}^{x} \left(2 + 2x + \frac{x^2}{2} - \frac{x^3}{6} - \right) dx$$

$$= 2 + 2x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{24}$$

Hence A is correct option

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SET-15
Numerical Methods

1. Solving of an engineering problem requires four steps. In order of sequence, the four steps are
   a) formulate, solve, interpret, implement
   b) solve, formulate, interpret, implement
   c) formulate, solve, implement, interpret
   d) formulate, implement, solve, interpret

2. The value of x that satisfies $f(x) = 0$ is called the
   a) root of an equation $f(x) = 0$
   b) root of a function $f(x)$
   c) zero of an equation $f(x) = 0$
   d) none of the above

3. A quadratic equation has _____ root(s).
   a) One
   b) Two
   c) Three
   d) four

4. An equation such as $\tan x = x$ has _____ root(s).
   a) Zero
   b) One
   c) Two
   d) infinite

5. The velocity of a body is given by $v(t) = 5e^{-t} + 4$, where t is in seconds and v is in m/s. The velocity of the body is 6 m/s at t = _____ seconds.
   a) 0.1823
   b) 0.3979
   c) 0.9163
   d) 1.609

6. For a certain cubic equation, at least one of the roots is known to be a complex root. How many total complex roots does the cubic equation have?
   a) One
   b) Two
   c) Three
   d) cannot be determined

7. The bisection method of finding roots of nonlinear equations falls under the category of a (an) _____ method.
   a) Open
   b) Bracketing
   c) Random
   d) Graphical

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8. If $f(x)$ is a real continuous function in $[a,b]$, and $f(a)f(b) < 0$, then for $f(x) = 0$, there is (are) _____ in the domain $[a,b]$.
   a) one root
   b) an undeterminable number of roots
   c) no root
   d) at least one root

9. The convergence of the bisection method is
   a) Linear
   b) Quadratic
   c) Cubic
   d) All of these

10. For an equation like $x^2 = 0$, a root exists at $x = 0$. The bisection method cannot be adopted to solve this equation in spite of the root existing at $x = 0$ because the function $f(x) = x^2$
   a) is a polynomial
   b) has repeated roots at $x = 0$
   c) is always non-negative
   d) has a slope equal to zero at $x = 0$

11. The second approximation using bisection method to the root of equation $x^3 - 4x - 9 = 0$ between 2 and 3 is
   a) 2.25
   b) 2.5
   c) 2.75
   d) 2.625

12. The Newton-Raphson method of finding roots of nonlinear equations falls under the category of _____ methods.
   a) Bracketing
   b) Open
   c) Random
   d) graphical

13. The Newton-Raphson method formula for finding the square root of a real number R from the equation $x^2 - R = 0$ is,
   a) $x_{i+1} = \frac{x_i}{2}$
   b) $x_{i+1} = \frac{3x_i}{2}$
   c) $x_{i+1} = \frac{1}{2}\left(x_i + \frac{R}{x_i}\right)$
   d) $x_{i+1} = \frac{1}{2}\left(3x_i - \frac{R}{x_i}\right)$

14. The next iterative value of the root of $x^2 - 4 = 0$ using the Newton-Raphson method, if the initial guess is 3, is
   a) 1.5
   b) 2.067
   c) 2.167
   d) 3.000

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15. The root of the equation $f(x)=0$ is found by using the Newton-Raphson method. The initial estimate of the root is $x_0=3$, $f(3)=5$. The angle the line tangent to the function $f(x)$ makes at $x=3$ is $57°$ with respect to the *x*-axis. The next estimate of the root, x_1 most nearly is
    a) –3.2470
    b) −0.2470
    c) 3.2470
    d) 6.2470

16. The secant method of finding roots of nonlinear equations falls under the category of _____ methods.
    a) Bracketing
    b) graphical
    c) open
    d) random

17. Find first approximation for to a real root of equation $x^2-3x-5=0$ by the method of false position between 2 and 3
    a) 3.400
    b) 3.667
    c) 3.500
    d) 3.625

18. Which of the following methods cannot be applied for locating complex roots of an equation
    a) Bisection method
    b) Regula falsi method
    c) Secant method
    d) Newton Raphson method

19. After newton raphson method which of the following has fastest rate of convergence
    a. Bisection method
    b. Regula falsi method
    c. Secant method
    d. Successive approximation method

20. For finding the root of $\sin x=0$ by the secant method, the following choice of initial guesses would not be appropriate.
    a) $\frac{\pi}{4}$ and $\frac{\pi}{2}$
    b) $\frac{\pi}{4}$ and $\frac{3\pi}{4}$
    c) $-\frac{\pi}{2}$ and $\frac{\pi}{2}$
    d) $\frac{\pi}{3}$ and $\frac{\pi}{2}$

21. Physically, integrating $\int_a^b f(x)dx$ means finding the
    a) Area under the curve from a to b
    b) Area to the left of point a
    c) Area to the right of point b
    d) Area above the curve from a to b

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22. The exact value of $\int_1^3 xe^x dx$ is most nearly
    a) 40.8036
    b) 40.1710
    c) 14.1034
    d) 19.611

23. The two-segment trapezoidal rule of integration is exact for integrating at most _____ order polynomials.
    a) first
    b) second
    c) third
    d) fourth

24. The value of $\int_1^3 xe^x dx$ by using the one-segment trapezoidal rule is most nearly
    a) 61.6722
    b) 61.8074
    c) 62.9748
    d) 64.6331

25. Simpson's formula is applicable to
    a) Odd number of intervals
    b) Even number of intervals
    c) Odd and even intervals
    d) None

26. The value of $\int_1^{1.5} x\text{^}2\ dx$ using Simpson's rule will be
    a) 1.3298
    b) 1.5201
    c) 1.5833
    d) 1.3277

27. The value of integral e^x is evaluated from 1 to 1.4 by using the following formulae. Which of these will give the least error ?
    a) Trapezoidal rule with h = 0.2
    b) Trapezoidal rule with h = 0.1
    c) Simpson's 1/3 rule with h = 0.1
    d) All of these

28. The results obtained by using Simpson's rule will be greater than those by using the trapezoidal rule
    a) In all cases
    b) Provided the intervals are small
    c) Provided the boundary is concave towards base line
    d) Provided the boundary is convex towards base line

29. The highest order of polynomial integrand for which Simpson's 1/3 rule of integration is exact is
    a) First
    b) Second
    c) Third
    d) Fourth

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30. The interpolating polynomial for Simpson's rule is
    a) Parabola
    b) Square
    c) Cubic curve
    d) Straight line

31. A differential equation is considered to be ordinary if it has
    a) one dependent variable
    b) more than one dependent variable
    c) one independent variable
    d) more than one independent variable

32. The differential equation $2\frac{dy}{dx} + x^2 y = 2x + 3,\ y(0) = 5$ is
    a) Linear
    b) nonlinear
    c) linear with fixed constants
    d) undeterminable to be linear or nonlinear

33. To solve the ordinary differential equation
    $3\frac{dy}{dx} + 5y^2 = \sin x,\ y(0) = 5$
    by Euler's method, you need to rewrite the equation as
    a) $\frac{dy}{dx} = \sin x - 5y^2,\ y(0) = 5$
    b) $\frac{dy}{dx} = \frac{1}{3}\left(\sin x - 5y^2\right),\ y(0) = 5$
    c) $\frac{dy}{dx} = \frac{1}{3}\left(-\cos x - \frac{5y^3}{3}\right),\ y(0) = 5$
    d) $\frac{dy}{dx} = \frac{1}{3}\sin x,\ y(0) = 5$

34. Given $3\frac{dy}{dx} + \sqrt{y} = e^{0.1x},\ y(0.3) = 5$ and using a step size of $h = 0.3$, the best estimate of $\frac{dy}{dx}(0.6)$ using Euler's method is most nearly
    a) -3.7319
    b) 3.6288
    c) 4.8964
    d) -4.8964

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35. To solve the ordinary differential equation

$$1/3\frac{dy}{dx} + xy^2 = \cos x, y(0) = 2,$$

by Runge-Kutta 4th order method, you need to rewrite the equation as

a) $\frac{dy}{dx} = \cos x - xy^2, y(0) = 2$

b) $\frac{dy}{dx} = 3(\cos x - xy^2), y(0) = 2$

c) $\frac{dy}{dx} = \frac{1}{3}\left(-\cos x - \frac{xy^3}{3}\right), y(0) = 0$

d) $\frac{dy}{dx} = \frac{1}{3}\cos x - xy, y(0) = 2$

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Solutions:

1. Ans: a)

   The four steps of solving engineering problems are formulate the problem, solve problem using appropriate methods, interpret results and implement the results.

2. Ans: a)

   The root of an equation f(x) = 0 is defined as the point where f(x) is zero.

3. Ans: b)

   General quadratic equation is $ax^2 + bx + c = 0$. Therefore, it has two roots.

4. Ans: d)

   As tan 0 = 0 and tan 90 = ∞, therefore e can say that tan x = x has infinite roots.

5. Ans: c)

   $v(t) = 5e^{-t} + 4$

   We have v(t) = 6

   $\therefore\ 5e^{-t} + 4 = 6$

   $\therefore\ e^{-t} = \frac{2}{5}$

   $\text{In}\ e^{-t} = \text{In}\ \frac{2}{5}$

   $\therefore$ t = 0.9163 s.

6. Ans: b)

   There are two possible cases for the roots of a cubic equation.

   (1) Three real roots

   (2) One real root and two complex roots.

   Complex roots always come in a conjugate pair i.e. (a + ib) and (a – ib).

7. Ans: b)

   The bisection method is a bracketing method since it is based on finding the roots between two guesses that bracket the roots, i.e. where real continuous function f(x) in the equation changes the sign between the two guesses.

8. Ans: d)

   If f(a).f(b) < 0, then f(a) and f(b) have opposite signs. Since f(x) is continuous between a and b, the function needs to cross the x-axis. The point where the function f(x) crosses the x-axis is solution for f(x) = 0.

9. Ans: a)

   The convergence of bisection methods is linear.

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10. Ans: c)

We have $f(x) = x^2$

For bisection method, we need the f(x) for one of the roots. i.e. x must be negative. In case of $f(x) = x^2$, f(x) will never have negative values. i.e. f(x) will always be non-negative.

11. Ans: c)

$f(x) = x^3 - 4x - 9$.

$a = 2, b = 3$

$f(a) = 8 - 8 - 9 = -9$

$f(b) = 27 - 12 - 9 = 6$

$\therefore$ roots lie between 2 and 3

Using bisection method,

$$a_1 = \frac{a+b}{2} = \frac{2+3}{2} = 2.5$$

$f(a_1) = 15.625 - 10 - 9 = -3.375$ is negative.

$\therefore$ root lies between a_1 and b

$$\therefore a_2 = \frac{a_1+b}{2} = \frac{2.5+3}{2} = \frac{5.5}{2} = 2.75$$

$\therefore$ $a_2 = 2.75$ is the answer.

12. Ans: b)

Newton - Raphson method is an open method since the guess of the root that is needed to get the iterative process started is a single point. Other open methods such as the secant method use two initial guesses of the root, but they do not have to bracket the root.

13. Ans: c)

By Newton - Raphson method

$$x_{i+1} = x_i - \frac{f(x_i)}{f^1(x_i)}$$

$f(x) = x^2 - R$

$f^1(x) = 2x$

$$x_{i+1} = x_i - \frac{x_i^2 - R}{2x_i} = x_i - \frac{x_i}{2} + \frac{R}{2x_i}$$

$$x_{i+1} = \frac{1}{2}x_i + \frac{1}{2}\frac{R}{x_i}$$

14. Ans: c)

$f(x) = x^2 - 4$, $x_1 = 3$

$f^1(x) = 2x$

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$$x_{i+1} = x_i - \frac{f(x_i)}{f^1(x_i)}$$

$f(3) = 9 - 4 = 5$

$f^1(3) = 6$

$$x_2 = 3 - \frac{5}{6} = \frac{13}{6}$$

$x_2 = 2.167$

15. Ans: c)

$x_1 = 0$

$f(x) = \cos x - 3x + 1.$

$f(0) = 1 + 1 = 2$

$f^1(x) = -\sin x - 3$

$f^1(0) = -3$

$$\therefore x_2 = 0 - \frac{2}{-3} = \frac{2}{3} = 0.67$$

16. Ans: c)

Secant method of finding roots of non-linear equations falls under the category of open methods.

17. Ans: a)

By false position method

$$x_i = \frac{a.f(b) - b.f(a)}{f(b) - f(a)}$$

$f(x) = x^2 - 3x - 5$; $a = 2$; $b = 3$

$f(a) = 4 - 6 - 5 = -7$

$f(b) = 9 - 6 - 5 = -2$

$$x_i = \frac{2(-2) - 3(-7)}{-2 - (-7)} = \frac{-4 + 21}{5} = \frac{17}{5}$$

$x_i = 3.40$

18. Ans: a)

Bisection cannot be applied for locating complex roots of an equation. It is applicable only to equations having real roots.

19. Ans: c)

The order of convergence for Newton – Raphson is 2, whereas order of convergence for secant methods is 1.62, and bisection as 1.

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20. Ans: b)

$\frac{\pi}{4}$ and $\frac{3\pi}{4}$ are not good initial guesses because $\sin\frac{\pi}{4} = \sin\frac{3\pi}{4} = \frac{1}{\sqrt{2}}$ which would result in division by 0 in formula – which is not defined.

21. Ans: a)

Integration is nothing but area under the curve.

22. Ans: b)

$\int_1^3 xe^x dx$ is most nearly 40.171 by using integration by parts rule.

23. Ans: a)

The single trapezoidal rule is exact for atmost a first order polynomial. The two segment is also exact for a first order polynomial only.

24. Ans: c)

$\int_1^3 xe^x dx$

By trapezoidal rule

$$\int_a^b f(x)\,dx \approx (b-a)\left(\frac{f(a)+f(b)}{2}\right)$$

Here, a = 1 and b = 3

$f(x) = x.e^x$

$f(a) = 1.\ e^1 = e = 2.7182$

$f(b) = 3.\ e^3 = 60.2566$

$$\int_a^b f(x)\,dx \approx (3-1)\left(\frac{2.7182+60.2566}{2}\right) \approx 60.9748$$

25. Ans: c)

Simpson's formula is applicable to both even and odd intervals.

26. Ans: c)

a = 1, b = 1.5 and h = $\frac{1.5-1}{1} = 0.5$

$f(x) = x^2$

f(a) = 1 and f(b) = 2.25

$$f\left(\frac{a+b}{2}\right) = 1.5625$$

By Simpson's one-third rule

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$$\int_a^b f(x) \approx \frac{h}{3}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

$$= \frac{0.5}{3}(1 + 4 \times 1.5625 + 2.25)$$

$$= 1.583$$

27. Ans: c)

The error is reduced by using Simpson's rule.

28. Ans: c)

The results obtained by Simpson's rule are greater wherever boundary is concave towards base line.

29. Ans: c)

In Simpson's rule, the highest order of the polynomial integrand is 3.

30. Ans: c)

The interpolating polynomial of order-3 is a cubic curve.

31. Ans: c)

The differential equation is considered ordinary if it has one independent variable.

32. Ans: a)

A differential equation is linear if the coefficients of the dependent variable and all the derivatives of the dependent variable in the ODE are only functions of the independent variable.

33. Ans: b)

Euler's method is $\frac{dy}{dx} = f(x)$

$\therefore\ 3\ \frac{dy}{dx} + 5y^2 = \sin x$

$3\ \frac{dy}{dx} = \sin x - 5y^2$

$\frac{dy}{dx} = \frac{1}{3}\left(\sin x - 5y^2\right)$; y (0) = 5

34. Ans: c)

$$\frac{dy}{dx} = \frac{1}{3}\left(e^{01.x} - \sqrt{y}\right)$$

$x_0 = 0.3$, $y_0 = 5$, $h = 0.3$, $x_1 = 0.6$

$y_1 = 5 + \frac{1}{3}\left(e^{0.03} - \sqrt{5}\right) \times 0.3 = 5 + (1.03 - 2.236) \times 0.1$

$y_1 = 5 + (-0.1036) = 4.8964$

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35. Ans: b)

$$\frac{1}{3}\frac{dy}{dx} + xy^2 = \cos x \text{ , } y(0) = 2$$

$$\frac{1}{3}\frac{dy}{dx} = \cos x - xy^2 \text{ , } y(0) = 2$$

$$\frac{dy}{dx} = 3\left(\cos x - xy^2\right) \text{ , } y(0) = 2$$

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Past GATE Questions

1. Match the following and choose the correct combination:
   Group-I
   A. Newton-Raphson method
   B. Runge-kutta method
   C. Simpson's Rule
   D. Gauss elimination

   Group-II

   1. Solving nonlinear equations
   2. Solving linear simultaneous equations
   3. Solving ordinary differential equations
   4. Numerical Integration
   5. Calculation of Eigen values

(G-2005, 1-Marks)

2. For the function e^{-x} , the linear approximation around $x = 2$ is

a) (3-x)e^{-2}

b) 1-x

c) $(3 + 2\sqrt{2} - (1 + \sqrt{2}x)e^{-2}$

d) e^{-2}

(G-2007, 1-Marks)

3. The equation $x^3 - x^2 + 4x - 4 = 0$ is to be solved using the Newton – Raphson method. If $x = 2$ is taken as the initial approximation of the solution, then the next approximation using this method will be

a) $\frac{2}{3}$
b) $\frac{4}{3}$
c) $\mathbf{1}$
d) $\frac{3}{2}$

(G-2007, 2-Marks)

4. The recursion relation to solve $x = e^{-x}$ using Newton-Raphson method is

a) $x_{n+1} = e^{-x_n}$
b) $x_{n+1} = x_n - e^{-x_n}$
c) $x_{n+1} = (1 + x_n)\frac{e^{-x_n}}{1 + e^{-x_n}}$
d) $x_{n+1} = {x_n}^2(x_n)\frac{e^{-x_n}(1+x_n)-1}{x_n + e^{-x_n}}$

(G-2008, 2-Marks)

5. Consider the differential equation $\frac{dy(x)}{dx} - y(x) = x$ with the initial condition $y(0) = 0$. Using Euler's first order method with a step size of 0.1, the value of $y(0.3)$ is
   a) 0.01

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b) 0.031
c) 0.0631
d) 0.1

(G-2010, 2- Marks)

6. A numerical solution of the equation $f(x) = x + \sqrt{x} - 3 = 0$ can be obtained using Newton-Raphson method . If the starting value is x=2 for the iteration, the value of x that is to be used in the next step is
   a) 0.306
   b) 0.739
   c) 1.694
   d) 2.306

   (G-2011, 2-Marks)

7. The maximum value of θ until which the approximation $\sin\theta = \theta$ holds to with in 10 % error is
   a) 10°
   b) 18°
   c) 50°
   d) 90° **(G-2013, 1-Marks)**

8. A Polynomial $f(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x - a_0$ with all coefficients positive has
   a) *no real roots*
   b) *no negative real roots*
   c) *odd number of real roots*
   d) *at least one positive and one negative real root* **(G-2013, 1-Marks)**

9. Match the application to appropriate numerical method. **(G-2014, 1- Marks)**

Application	**Numerical methods**
P1 : Numerical Integration	**M1:** Newton-Raphson Method
P2 : Solution to a transcendental equation	**M2:** Runge-Kutta Method
P3: Solution to a system of linear equations	**M3:** Simpson's 1/3 rule
P4: Solution to a differential equation	**M4:** Guass Elimination method

   a) P1 – M3, P2-M2, P3 – M4, P4- M1
   b) P1 – M3, P2-M1, P3 – M4, P4- M2
   c) P1 – M4, P2-M1, P3 – M3, P4- M2
   d) P1 – M2, P2-M1, P3 – M3, P4- M4

10. The Newton-Raphson method is used to solved the equation $f(x) = x^3 - 5x^2 + 6x - 8 = 0$

    Taking the initial guess as $x = 5$, the solution obtained at the end of the first iteration is ……

    (G-2015, 2-Mark)

11. Consider the first order initial value problem $\dot{y} = y + 2x - x^2, y(0) = 1, (0 \le x \ge \infty)$

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With exact solution $(x) = x^2 + e^x$. For $x = 0.1$, the percentage difference between the exact solution and the solution obtained using a single iteration of the second-order Runge-Kutta method with step-size $h = 0.1$ is …..

(G-2016, 1-Mark)

12. The ordinary differential equation $\frac{dx}{dt} = -3x + 2\,,\ with\ x(0) = 1$

is to be solved using the forward Euler method.

The largest time step that can be used to solve the equation without making the numerical solution unstable is …..

(G-2016, 2-Mark)

13. Starting with $x = 1$, the solution of the eqaution $x^3 + x = 1$ after two iterations of Newton-Raphson's method (upto two decimal places) is……..

(G-2017, 1-Mark)

Answers

1)c 2) a 3)b 4) c 5)b 6) c 7) b 8) d 9) b

Solutions

1. **Ans. c**
2. **Ans. a**
3. **Ans. b**

$Since\ it\ is\ given\ in\ the\ question\ f(x) = x^3 - x^2 + 4x - 4\ then\ f(2) = 8$

$and\ then\ f'(x) = 3x^2 - 2x + 4\ \ and\ f'(2) = 8$

By Newton-Raphson Method

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2 - \frac{8}{12} = \frac{4}{3}$$

4. **Ans . c**

It is given in the question that $x = e^{-x}$

Then we can write it as $f(x) = x - e^{-x} = 0\ then\ f'(x) = 1 + e^{-x}$

By Newton-Raphson Method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Here $f(x_n) = x_n - e^{-x_n}\ then\ f'(x_n) = 1 + e^{-x_n}$

And $x_{n+1} = x_n - \frac{x_n - e^{-x_n}}{1+e^{-x_n}} = (1 + x_n)\frac{e^{-x_n}}{1+e^{-x_n}}$

5. **Ans . b**

It is given that $\frac{dy}{dx} = x + y$

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x	y	$\frac{dy}{dx}$	$y' = y + (\frac{dy}{dx}) \times h$
0	0	0	0
0.1	0	0.1	0.01
0.2	0.01	0.21	0.031
0.3	0.031		

Then $y(0.3) = 0.031$

6. Ans. c

Since we know that, by Newton's Method, the $(n+1)^{th}$ Iterative root will be

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Here $x_0 = 2,\ then\ f(x_0) = \sqrt{2} - 1\ \ and\ f'(x_0) = 1 + \frac{1}{2\sqrt{2}}$

$x_1 = 2 - \frac{\sqrt{2}-1}{1+\frac{1}{2\sqrt{2}}} = 1.694$

7. Ans. b

Since, $10^o = \frac{10\,\pi}{180} = 0.1745\ \ and \sin 10^o = 0.1736, we\ can\ say\ that \sin\theta \cong \theta\ within\ 10\ \%\ error$

Similarly for $\theta = 18^o,\ \sin\theta \cong \theta\ within\ 10\ \%\ error$

But same is not true for $\theta = 50^o\ \ and\ \theta = 90^o$

Hence the maximum value of θ for the approximation $\sin\theta \cong \theta\ within\ 10\ \%\ error$ is18^o

8. Ans. d

9. Ans. b

10. Ans. 4.2903

$Since\ f(x) = x^3 - 5x^2 + 6x - 8\ , then\ f'(x) = 3x^2 - 10x + 6\ \ and\ \ x_0 = 5$

$then\ the\ next\ iterative\ root\ by\ Newtons\ method\ will\ be$

$$x_1 = 5 - \frac{f(5)}{f'(5)} = 5 - \frac{22}{31} = 4.2903$$

11. Ans. 0.06 %

$\frac{dy}{dx} = y + 2x - x^2\ ,\qquad y(0) = 1\ , f(x,y) = y + 2x - x^2$

For $x_0 = 0,\ y_0 = 1\ and\ h = 0.1\ ,\quad k_1 = h\,f(x_0, y_0) = 0.1$ and $k_2 = h\,f(x_0 + h,\ y_0 + k_1) = 0.129$

then $y_1 = y_0 + \frac{1}{2}(k_1 + k_2) = 1.1145$

Here exact solution will be $y(x) = x^2 + e^x = (0.1)^2 + e^{0.1} = 1.1152$

Then error is given by, $Error = 1.1152 - 1.1145 = 0.00069\ and\ \%\ error = 0.06\ \%$

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12. Ans. 0.66

$\frac{dy}{dx} = -3y + 2\,, \quad y(0) = 1,$

$if \ \ |1 - 3h| < 1, then\ solution\ of\ differential\ equation\ is\ stable \ \ ,$

$then - 1 < (1 - 3h) < 1 \quad and \quad 0 < h < \frac{2}{3}$ hence $h_{max} = \frac{2}{3} = 0.66$

13. Ans. 0.686

$f(x) = x^3 + x - 1\,, then\ f'(x) = 3x^2 + 1\ and\ x_0 = 1\,,\ f(1) =$ 1 and f'(1) = 4

Then by Newton's Method

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ for $x_0 = 1$, next iterative root is given by

$x_1 = 1 - \frac{1}{4} = 0.75 \quad and\ x_2 = 0.686$

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